The Aggregate Impact of Policy Uncertainty in the Green Energy Transition

Anna French*

November 2026

Click here for the latest version

Abstract

Federal wind and solar subsidies are central to U.S. electricity-sector decarbonization, yet they have been historically unstable. I develop a dynamic spatial general-equilibrium model of the U.S. economy and electricity sector with stochastic subsidies and irreversible electricity capital, and use it to quantify the cost of this instability. I compare counterfactual transition paths under a stable subsidy, available in every year, and under a budget-neutral unstable subsidy whose availability changes at dates firms do not know in advance. Under the stable subsidy, renewable capacity is 13% higher by 2070 and gas capacity is lower, avoiding $360 billion in climate damages and $46 billion in mortality damages from local air pollution. I find that the stable subsidy generates $292 billion in net benefits. The effects are largest in regions such as the Midwest, where the subsidy determines whether new renewable capacity is built. Perfect-foresight counterfactuals show that these gains come from eliminating policy switching itself; resolving uncertainty about the timing of policy changes while the subsidy still switches would lower welfare.

JEL Codes: D81, E22, H23, Q42, Q48, Q54, R13

Keywords: green energy transition, policy instability, policy uncertainty, irreversible investment, renewable subsidies, spatial general equilibrium

*Arizona State University, Department of Economics, asamodel@asu.edu. I am grateful to Wyatt Brooks for his guidance, encouragement, and support, and to Nicolai Kuminoff and Nicholas Vreugdenhil for their advice and many helpful discussions. I thank participants of the Job Market Workshop, PhD Seminar, and Macroeconomics Workshop at Arizona State University, the Modeling Climate Dynamics Workshop for Young Scholars at the University of Chicago, and the Arizona Workshop on Environment, Natural Resource, and Energy Economics at the University of Arizona for valuable feedback. I gratefully acknowledge support from the Washington Center for Equitable Growth. All errors are my own.

1 Introduction

The transition from fossil fuels to clean energy represents one of the largest reallocations of capital in history, with global clean-energy investment reaching $2.2 trillion annually (IEA, 2026). For decades, governments have used a range of climate policies to accelerate decarbonization. The majority of the macroeconomics literature that studies the effects of climate policies such as carbon taxes (Golosov et al., 2014), carbon tariffs (Casey et al., 2026a) and renewable subsidies (Arkolakis and Walsh, 2023) takes the future policy regime as given. In practice, developers make long-run investment decisions without knowing whether government subsidies will remain available in the future. The U.S. has subsidized wind through federal tax credits since 1992 and solar since 2006, but these subsidies have repeatedly changed:[1] the wind credit lapsed seven times between 1999 and 2022 and was each time renewed retroactively; the last renewal, the 2022 Inflation Reduction Act (IRA), extended federal clean-electricity tax credits for a decade, before the 2025 One Big Beautiful Bill Act (OBBBA) terminated the tax credits for new wind and solar projects. The aggregate cost of this renewable policy instability in the United States is unknown.

In this paper, I quantify the macroeconomic and environmental effects of renewable policy instability: federal subsidies whose availability changes at dates investors do not know in advance. I build a dynamic spatial general-equilibrium model of the U.S. economy and electricity sector with irreversible generation capital, convex adjustment costs, and a stochastic renewable subsidy. The model distinguishes generation capacity from electricity generation through merit-order dispatch, allowing changes in subsidy availability to affect emissions through both generation by the existing fossil fleet and investment in new fossil capacity. Electricity trade creates spatial spillovers: renewable investment in one region can displace fossil generation in another. I compute counterfactual transition paths under a stable subsidy, available in every year, and a budget-neutral unstable subsidy whose availability changes at dates regions do not know in advance. Under the stable subsidy, renewable capacity is 13% higher by 2070 than under the unstable subsidy, and less gas capacity is built. The additional fossil generation creates $360 billion in climate damages and $46 billion in local air-pollution mortality damages. At the same expected fiscal cost, the stable subsidy generates $292 billion in net benefits.

In the model, an unstable subsidy creates inefficiencies through two mechanisms. First, during a subsidy lapse, a developer can build without the subsidy or wait for the subsidy

[1] I refer to the federal production and investment tax credits as renewable subsidies throughout the paper.

to return. When the return date is uncertain, waiting has an option value: a project built during the lapse does not receive the subsidy, while one that waits can qualify when the subsidy returns. However, electricity demand continues to grow during policy lapses,[2] so fossil generation increases while renewable investment is delayed. Second, developers can bunch investment into periods in which the subsidy is available. Because convex adjustment costs make it costly to expand capacity rapidly, concentrating investment within eligibility windows creates additional resource costs. A stable subsidy removes the incentives to wait for subsidy renewal or to bunch investment into periods when the subsidy is available.

Importantly, delaying renewable investment does not require new fossil plants to be built for emissions to rise. Most of the emissions difference in the counterfactuals comes from greater generation by the existing fossil fleet rather than investment in new fossil capacity. While renewable investment is delayed, existing fossil plants generate more: higher utilization of existing coal and gas accounts for 80% of the cumulative emissions difference; the remaining 20% comes from additional gas capacity that would not have been built under the stable subsidy. Renewable investment can resume when the subsidy returns, but the additional gas capacity is irreversible and can continue to generate for decades. The emissions effect of an unstable subsidy therefore depends not only on which technologies are built, but also on how the existing electricity generation capacity is dispatched.

The aggregate results conceal substantial regional heterogeneity. A stable subsidy has little effect where renewable projects are profitable without the federal subsidy or remain unprofitable even with the subsidy. The effects are largest in marginal regions where the federal subsidy changes the renewable investment decision and existing fossil capacity can increase generation when investment is delayed. The Midwest combines both features: under the stable subsidy it builds the most additional renewable capacity, while under the unstable subsidy its existing coal and gas fleet generates more. I value the local air-pollution consequences of these generation changes using plant-specific marginal damages from AP4. Because coal has substantially higher mortality damages per MWh than gas, the Midwest accounts for more than half of the $46 billion in avoided local air-pollution mortality damages.

Renewable suitability also differs substantially across the U.S. regions. This means that the same renewable capacity can produce different amounts of electricity depending on where it is built. I model electricity trade with transmission constraints. Renewable resources and

[2]The U.S. Energy Information Administration projects U.S. electricity demand to grow by roughly 0.9–1.6% per year through 2050, driven in part by data centers and the electrification of transportation and buildings.

transmission costs therefore jointly determine where new capacity is built and where its electricity is consumed. Trade also creates spatial spillovers, as renewable investment in one region can reduce fossil generation elsewhere. As a result, California, for example, reduces gas generation partly by importing additional renewable electricity from other regions in the Western Interconnection.

I motivate the model with evidence from the history of U.S. renewable tax credits. I show that the production tax credit is large enough to affect whether wind projects are profitable, and that wind investment historically adjusts around changes in subsidy eligibility. For example, following the 2012 expiration deadline, annual additions fell from 13 GW to 1 GW. I also use renewable-equity prices to test whether recent changes in federal subsidies were known in advance. Renewable valuations respond to the IRA agreement, the 2024 election, House passage of the OBBBA repeal, and the Senate's subsequent revision, while fossil-energy equities do not change significantly. I use this suggestive evidence to build a model in which subsidy availability affects renewable investment and investors form expectations over future policy changes without knowing their realized dates in advance.

I calibrate the model using U.S. electricity data from the U.S. Energy Information Administration and the political history of the subsidy. The model has 13 electricity regions that differ in renewable resources, demand, initial generation capacity, and neighbors with whom they can trade electricity. Renewable and gas capital costs, electricity demand, fuel costs, capacity factors, and regional renewable resources are taken from external data and are exogenous in the model; investment-cost parameters are calibrated to national wind and solar capacity growth. The stochastic subsidy process is calibrated separately using observed transitions in unified federal control. The model reproduces untargeted aggregate emissions, wholesale electricity prices, and the generation mix.

I also use perfect-foresight counterfactuals to separate the effects of policy switching itself from uncertainty about when policy changes occur. The subsidy follows the same realized switching schedules, but firms learn the entire future schedule in 2028. Knowing the dates does not eliminate the value of a stable subsidy: moving from the switching subsidy under perfect foresight to the stable subsidy is worth \$355 billion. In contrast, the value of resolving uncertainty about the timing is negative, −\$63 billion. With perfect foresight, firms concentrate investment more strongly around known subsidy windows, and convex adjustment costs make this additional bunching costly. Most of the welfare loss therefore comes from policy switching itself; uncertainty about the exact dates of policy changes partially mitigates it.

The results are robust to the main uncertainty in the policy calibration. The duration of a future subsidy lapse is not identified by the historical tax-credit record, because past lapses were short and ultimately reversed. I therefore treat the central calibration as a data-calibrated benchmark and bracket it with alternative forward regimes ranging from a rapidly returning subsidy to permanent repeal. The value of a stable subsidy remains positive throughout, ranging from $56 billion to $385 billion. Excluding local air-pollution damages, the corresponding range is $48 billion to $315 billion. The result also remains positive when climate damages are discounted at the private market rate.

The model abstracts from several margins that could affect the magnitude and composition of the response. Renewable capital costs and coal retirements are exogenous, and the model does not include storage or state clean-energy policies. Section 7.4 discusses these and other omitted mechanisms.

More broadly, the results show that changes in policy can create substantial costs during the green energy transition even when expected government spending is unchanged. When investment is irreversible and costly to shift across time, changes in policy availability affect when firms invest, which technologies are built, and how the existing capital stock is used while investment is delayed. Regional heterogeneity and costly electricity trade create spatial spillovers, so changes in renewable investment in one region can affect production and environmental damages elsewhere. The value of stable climate policy therefore comes from providing continuity during the transition rather than from reducing uncertainty about the dates of future policy changes.

1.1 Related literature

This paper contributes to the literature on policy uncertainty and irreversible investment. Uncertainty creates an option to wait and can delay investment (Bernanke, 1983; Dixit and Pindyck, 1994) in the models with sunk costs. Expectations about future policy change can also affect investment before a policy change occurs: Lemoine (2017) shows that anticipated carbon pricing changes current energy investment, while French and Vreugdenhil (2026) show that firms acquire permits before anticipated regulation to preserve future investment opportunities. In electricity markets, Gowrisankaran et al. (2025) study how uncertainty over environmental regulation affects investment in coal generation, and Fried et al. (2022) quantify the macroeconomic effects of uncertainty over whether a carbon tax will be adopted. In my paper, I model both anticipatory and waiting mechanisms. I build on Barradale (2010),

who studies uncertainty surrounding repeated expirations of the U.S. production tax credit and argues that uncertainty about renewal can delay wind investment, and Chen (2025), who shows that anticipated expiration accelerates investment before eligibility ends. My model contains both margins: developers can wait during a subsidy lapse and shift investment toward periods in which the subsidy is available. I also separate uncertainty about future policy dates from the effects of policy switching itself: even under perfect foresight, known subsidy lapses change investment timing.

A related literature studies the design of clean-energy subsidies. Langer and Lemoine (2022) characterize optimal dynamic adoption subsidies, Aldy et al. (2023) compare investment and production subsidies for wind generation, and Ricks and Kay (2025) study the design of time-limited renewable subsidies. Aldy et al. (2026) examine the role of federal tax policy in U.S. power-sector decarbonization, while Casey et al. (2026b) study the macroeconomic and emissions effects of clean-energy subsidies. I focus on how the stability of federal subsidies over time affects the energy transition and compute the welfare effects in the general equilibrium.

The paper also contributes to work on the regional effects of policy uncertainty. Poilly and Tripier (2025) show that uncertainty about a common national trade policy has heterogeneous effects across U.S. regions because regions differ in their exposure to trade, while Falk and Shelton (2018) show that policy uncertainty can redirect investment across locations. In my setting, regions face the same federal subsidy process, but its effects are heterogeneous because regions differ in renewable resources and existing generation fleets. Electricity trade also creates spillovers across regions: renewable investment in one region can displace fossil generation in another. Because fossil technologies and local air-pollution damages differ across regions, the environmental value of renewable investment depends partly on where the fossil generation it displaces is located.

Finally, this paper contributes to quantitative general-equilibrium work on climate policy and the energy transition (Golosov et al., 2014; Acemoglu et al., 2012, 2016; Fried, 2018; Arkolakis and Walsh, 2023). Fried (2018) studies how climate policy redirects innovation toward clean technologies, while Acemoglu et al. (2023) show how the availability of natural gas can redirect innovation and alter the long-run energy transition. I take technology costs decline as given and focus on changes in the timing of irreversible generation investment and in the utilization of the existing generation stock: distinguishing generation capacity from electricity generation allows policy to affect environmental damages through both new investment and fossil dispatch.

Roadmap. The rest of the paper is organized as follows. Section 2 describes the institutional background. Section 3 presents empirical evidence on whether subsidy availability affects investment decision and whether the changes in subsidy were foreseen. Section 4 sets up the model and Section 5 calibrates it. Section 6 describes the counterfactuals, Section 7 reports the results, and Section 8 concludes.

2 Institutional background

The U.S. electricity sector has undergone a substantial change in its generation mix over the past two decades and this transition had two margins: switching from coal to natural gas, and then switching from natural gas to renewables. Coal was the leading source of electricity generation in the majority of states in 2001, but by 2025 natural gas had displaced it in most states, while wind had become the leading source in five states (Figure 1). Over the same period, wind and solar installation costs fell substantially, by 55% and 83% since 2010, respectively. Renewable capacity has consequently expanded rapidly, although fossil fuels continue to supply a large share of U.S. electricity.

The transition is geographically uneven because renewable suitability differs across the regions. Figure 2 reports wind and solar capacity factors[3] across the 13 electricity regions used in the model. Wind is especially suitable for the interior regions and Texas, while solar resources are strongest in the Southwest and California. These differences affect the cost of renewable electricity and therefore where wind and solar capacity is built. They also make the effect of a common federal subsidy heterogeneous across space and can create winners and losers.

The Federal Government uses separate tax credits for wind and solar: the Production Tax Credit for wind (a per-MWh payment), and the Investment Tax Credit for solar (a fixed share of capital cost). Both are claimed under a start-of-construction provision that grandfathers the subsidy at the rate that exists when a project breaks ground[4]. The wind PTC lapsed seven times between 1999 and 2022; each lapse was a congressional delay and was renewed retroactively. A wind project that qualifies while the PTC is available retains the

[3]The capacity factor is the ratio of actual electricity generation to the amount a facility would generate if it operated at nameplate capacity continuously.

[4]The wind production tax credit had a placed-in-service rule: a project qualified only if it reached commercial operation before the credit expired, until the American Taxpayer Relief Act (January 2013) replaced it with the start-of-construction rule.

credit over its eligibility window[5] so projects that have already qualified remain insulated from subsequent changes in eligibility. Solar subsidies were stable until 2025. The 2022 Inflation Reduction Act committed both subsidies for a decade. The 2025 One Big Beautiful Bill Act then repealed the subsidies for projects placed in service after 2027, the first full subsidy reversal in their history. (Figure 13) shows the full history of the subsidies.

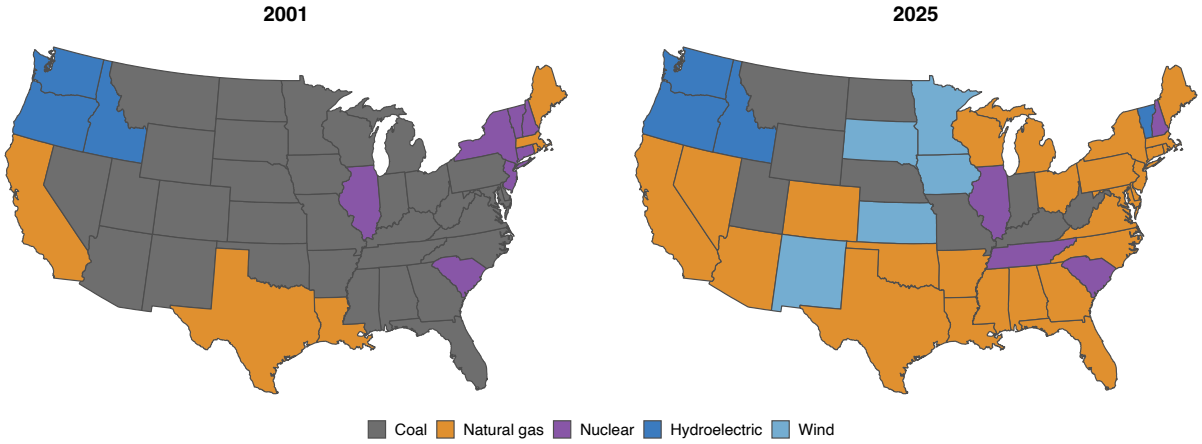


Figure 1: The top source of electricity generation by state, 2001 and 2025.
Notes. Sources: EIA Electric Power Annual, net generation by state and energy source (2001), and EIA-923 (2025).

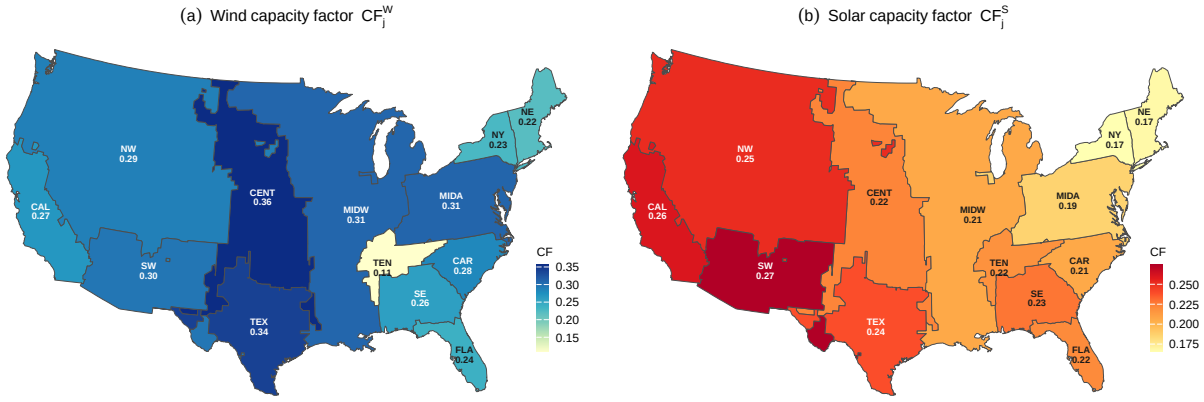


Figure 2: Wind and solar capacity factors across the 13 EIA regions.
Notes. Renewable resource geography across the 13 EIA electricity regions. *Panel (a):* wind (CF_j^W). *Panel (b):* solar (CF_j^S). Source: net generation (EIA-923) over nameplate capacity (EIA-860), 2019–2024.

For most of its history, subsidy availability did not depend on political party in power. Earlier lapses were short legislative interruptions followed by renewal, and the credit sur-

[5]For the PTC, an eligible wind facility generally receives the credit on qualifying generation during its first ten years of operation.

vived periods of unified Republican government in 2003–2006 and 2017–2018. Congressional support subsequently became increasingly polarized: Republican support for legislation containing the credit declined from roughly 75% in 1992 to zero, while Democratic support remained high. After the IRA, the credits became a partisan target, and OBBBA reversed subsidies under unified Republican control. The partisan divide also appears in public opinion: in a 2025 University of Chicago Energy Policy Institute survey, 73% of Democrats but 39% of Republicans described expanding the U.S. clean-energy industry as extremely or very important for the federal government (Energy Policy Institute at the University of Chicago and AP-NORC, 2025).

These changes motivate modeling the relationship between future federal subsidy availability and changes in political control. Section 5 describes how I use the historical political record to calibrate the stochastic policy process.

3 Empirical evidence

I use two facts to justify the model's assumptions about policy uncertainty. First, I show that the federal wind subsidy is large enough to affect project profitability and that wind investment adjusted around changes in subsidy eligibility in the last 25 years. Second, I use financial-market responses to recent federal policy events to test whether investors knew the realized future policy path in advance.

3.1 The subsidy availability affects wind investment

The production tax credit is large relative to wind-project revenue. A wind plant earns the wholesale price of electricity and, during its first ten years of eligibility, the §45 production tax credit of $26/MWh for each megawatt-hour it generates. I combine EIA-860 and EIA-923 data with engineering cost assumptions to calculate project revenue and levelized cost[6] across the observed distribution of wind capacity factors. At the $50/MWh average on-peak wholesale price over 2010–2024 across 8 trading hubs (EIA/ICE), the PTC accounts for 34% of project revenue during the years in which it is paid. The representative project's levelized cost is $53/MWh, slightly above the wholesale price of $50/MWh without the subsidy.

[6]The levelized cost of electricity is the average cost per MWh of generating electricity over a plant's lifetime, including its up-front capital cost and operating costs.

Without the credit, the representative wind project breaks even at a capacity factor of 0.35; the subsidy lowers the breakeven capacity factor to 0.23. Of the 129 GW of operating U.S. wind capacity in the calculation, 48% lies above the unsubsidized breakeven threshold, 35% lies between the subsidized and unsubsidized thresholds, and the remaining 17% lies below the subsidized threshold. Thus, for a substantial share of the observed wind-capacity distribution, the PTC is large enough to move the representative project across the profitability threshold.

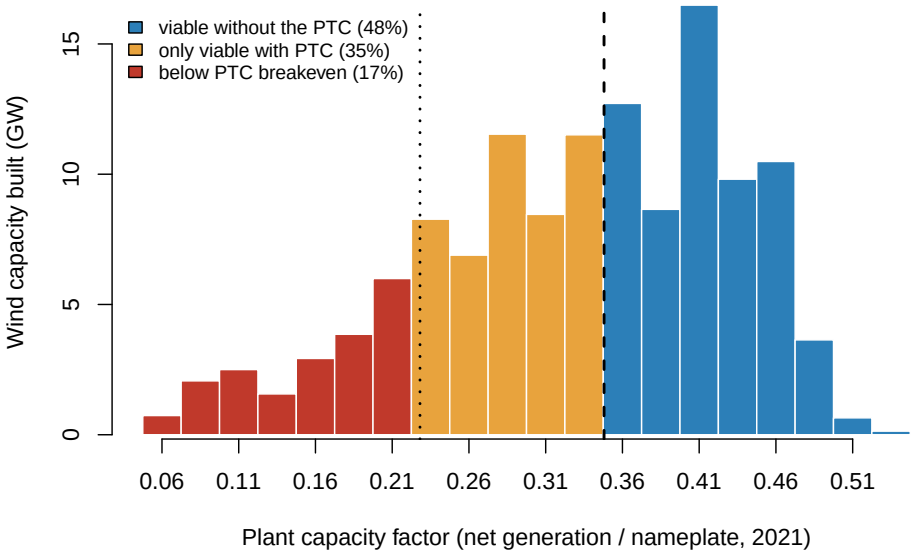


Figure 3: Wind capacity across subsidized and unsubsidized breakeven thresholds.

Notes. Operating U.S. wind capacity (EIA-860, 2021) by realized capacity factor (net generation, EIA-923, over nameplate capacity). The dashed line is the capacity factor at which the representative project breaks even at the $50/MWh wholesale price (0.35); the dotted line is the breakeven capacity factor when the $26/MWh credit is added (0.23). Capacity between the two lines lies between the subsidized and unsubsidized breakeven thresholds under the assumptions of the calculation. Engineering assumptions are reported in Table 12. Sources: EIA-860 and EIA-923.

Wind investment also shows pronounced boom-bust patterns around PTC expiration dates. Annual capacity additions rose before historical deadlines and fell sharply afterward. After four of these lapses, annual wind additions fell in 2000, 2002, 2004, and 2013, by 79%, 60%, 71%, and 92%, respectively (Figure 4). The largest decline followed the 2012 deadline, when additions fell from 13 GW in 2012 to 1 GW in 2013. This investment pattern around the PTC has also been documented by Barradale (2010) and Chen (2025).

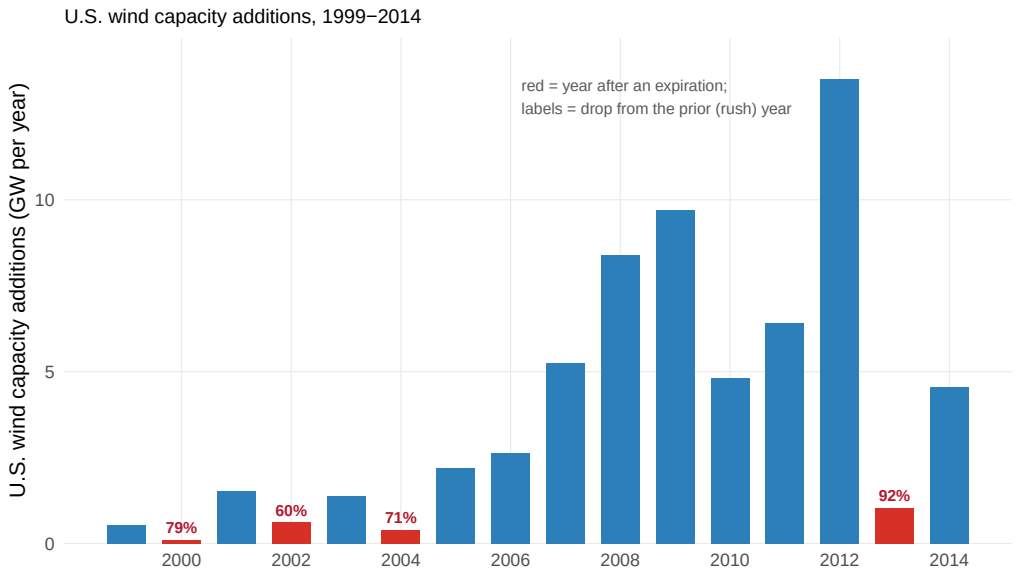


Figure 4: Wind capacity additions around four PTC expirations.

Notes. National U.S. wind capacity additions, 1999–2014. Bars for the four years following the deadlines (2000, 2002, 2004, and 2013) are shown in red and labeled with the decline from the preceding year. Source: EIA-860.

3.2 Evidence on investors' information about future policy

The second set of evidence explores investors' information about future policy. In the model, investors have rational expectations over future subsidy states and can anticipate the possibility of a policy change, but they do not know the realized future policy path. I use financial-market responses to justify this assumption.

Under full foresight, the realization of the event contains no new information and therefore should not change firm valuations. I consider four events: the Inflation Reduction Act (July 2022), the 2024 election, House passage of the OBBBA repeal (May 2025), and the Senate revision (June 2025). Figure 5 shows the changes for renewable- and fossil-energy equities. I calculate cumulative returns in excess of the S&P 500 from the event day through three trading days after the event. For each event, I summarize the differential response as

$$\Delta_e = R_e^{\text{ren}} - R_e^{\text{fos}}, \tag{3.1}$$

where R_e^{ren} and R_e^{fos} are cumulative abnormal returns for the renewable and fossil portfolios. I also report the same difference over the five trading days before each event, $[-5, -1]$, to show the extent of pre-event movement.

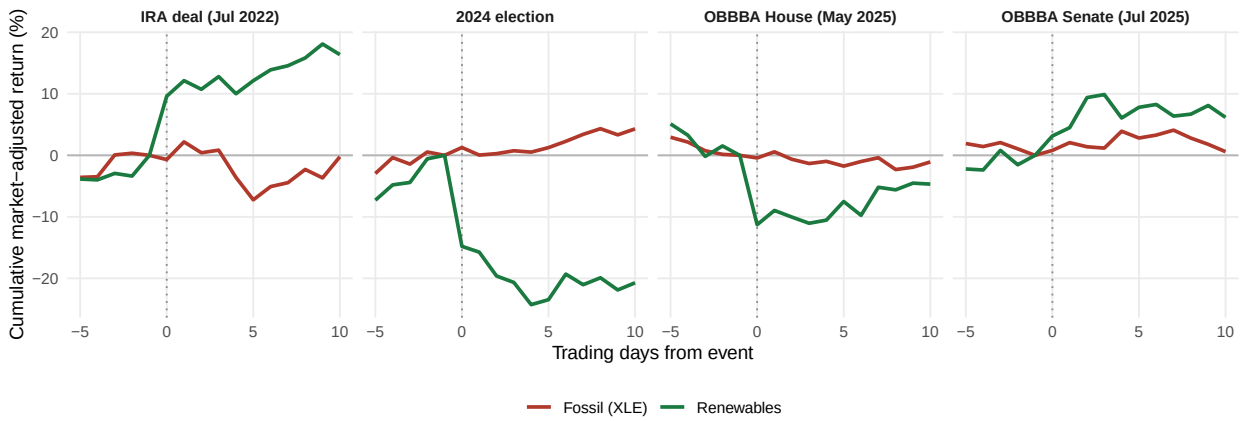


Figure 5: Stock-price reaction to federal policy events.

Notes. Cumulative return in excess of the S&P 500 normalized to zero on the trading day before the event. Source: daily stock prices.

Table 1: Renewable and fossil equity around federal policy events: cumulative abnormal returns.

Event	Renew.	Fossil	Difference $[0,+3]$		Placebo $[-5,-1]$	
	CAR (%)	CAR (%)	$R-F$ (pp)	t	$R-F$ (pp)	t
IRA announcement (Jul 2022)	+12.8	+0.8	+12.0	1.8	+2.2	0.3
2024 election (Nov 2024)	-20.7	+0.7	-21.4	-4.4	+3.9	0.7
OBBBA House (May 2025)	-11.0	-1.3	-9.7	-2.4	-1.7	-0.4
OBBBA Senate (Jul 2025)	+9.9	+1.2	+8.7	1.8	+11.1	2.1

Notes. CAR: cumulative abnormal return relative to the S&P 500. $R-F$ is the difference between the renewable and fossil portfolio CARs, in percentage points. The event window $[0,+3]$ starts from the event day through three trading days after the event. The placebo window $[-5,-1]$ reports the same renewable-minus-fossil difference over the five trading days preceding the event. The renewable portfolio includes TAN, FAN, ICLN, First Solar, Sunrun, Enphase, and NextEra; the fossil portfolio is XLE.

Renewable equities gained 12.8% following the IRA agreement, lost 20.7% after the 2024 election and 11.0% after House passage of the OBBBA repeal, and gained 9.9% when the Senate subsequently softened the repeal (Table 1 and Figure 5). The fossil equities did not change to the same extent. The signs of the renewable-minus-fossil differences track whether an event increased or reduced expected federal subsidies. The pre-event differences are generally smaller, although the Senate event shows substantial movement before the event window, so I do not interpret these estimates as causal event-study effects. I use this exercise as suggestive evidence to motivate the uncertainty in the model: investors can anticipate the possibility of policy changes, but they do not know the realized future subsidy path in advance.

4 Model

I develop a dynamic spatial general-equilibrium model that links stochastic renewable subsidies to investment, electricity generation, and emissions. The economy consists of 13 electricity regions connected through three interconnections. Households own generation capital and make irreversible investment decisions in wind, solar, and gas under uncertainty about future federal subsidies. Final-good firms use capital, labor, and electricity, and production capital is traded nationally. Within each region, installed generation capacity is dispatched in merit order against fluctuating electricity demand, and regions trade electricity within their interconnection. A capacity-adequacy requirement ensures that each region has enough reliable generation capacity to meet forecast demand.

4.1 Environment

Time is discrete and the horizon is infinite, $t = 2010, 2011, \ldots$. There are $J = 13$ regions indexed by j, corresponding to the Energy Information Administration electricity regions and grouped into the Eastern, Western, and Texas interconnections.[7] Each region has a representative household-generator that consumes, supplies labor inelastically, saves in a national capital market, and owns and invests in the region's electricity-generation capital, and a representative firm that produces the final good. Regions differ in wind and solar resources (Figure 2), electricity demand, initial generation capacity, and the interconnection within which they trade.

The economy transitions from observed 2010 capital stocks toward a stochastic steady state. The transition is driven by growing electricity demand, declining global technology costs, and federal renewable subsidies.

The federal subsidy is the model's only source of aggregate uncertainty. Subsidy availability follows a two-state Markov process $s_t \in \{0, 1\}$, where $s_t = 1$ is "subsidy on" and $s_t = 0$ is "subsidy off". Household-generators know the transition probabilities but not the dates of future policy changes.

Wind earns a production tax credit (PTC) of ϕ^{PTC} per MWh. Solar gets an investment

[7] The 13 regions are California (CAL), Carolinas (CAR), Central (CENT), Florida (FLA), Mid-Atlantic (MIDA), Midwest (MIDW), New England (NE), Northwest (NW), New York (NY), Southeast (SE), Southwest (SW), Tennessee (TEN), and Texas (TEX).

tax credit (ITC), ϕ^{ITC}, that reduces its up-front capital cost when the subsidy is active. Appendix A states the eligibility rules formally.

4.2 Households and generation investment

The representative household in region j consumes a final good C_{jt} and electricity E^{HH}_{jt}, supplies labor inelastically, saves in the national capital market, and owns the region's generation capital. It maximizes

$$\mathbb{E}_0 \sum_t \beta^t \log \mathcal{U}(C_{jt}, E^{HH}_{jt}), \qquad \mathcal{U}(C,E) = \left[C^{\frac{\sigma-1}{\sigma}} + \bar{\alpha}_h^{1/\sigma} E^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \tag{4.1}$$

where σ is the electricity-demand elasticity and $\bar{\alpha}_h$ is the electricity weight. I set σ using estimates from the empirical electricity-demand literature and, conditional on σ, calibrate $\bar{\alpha}_h$ so that electricity accounts for its observed 2% expenditure share (Section 5).

The household budget constraint is

$$C_{jt} + p^E_{jt} E^{HH}_{jt} + \sum_k \tilde{p}^k_t I^k_{jt} + \sum_k \Psi^k(I^k_{jt}, K^k_{jt}) + S_{jt} = (1-\varsigma_t) w_{jt} L_{jt} + r^K_t A_{jt} + \Pi^E_{jt} + \mathcal{R}_{jt} + N_j T^{nat}_t, \tag{4.2}$$

where S_{jt} is saving in the national capital market, Π^E_{jt} is electricity-sector profit, $\mathcal{R}_{jt}$ is the region's lump-sum share of interregional congestion revenues, and T^{nat}_t is a national lump-sum transfer. Financial wealth evolves according to

$$A_{j,t+1} = (1-\delta_P) A_{jt} + S_{jt}.$$

The labor-income tax ς_t and the lump-sum transfer T^{nat}_t finance federal subsidy payments, and the federal budget balances each period. In the consolidated household–government budget the tax cancels algebraically, leaving the household's effective budget in full resource prices (Appendix A.5); this cancellation holds for any labor-supply elasticity. Because labor is supplied inelastically, the tax moreover distorts no margin, so ς_t is also immaterial for allocations and welfare. The subsidy affects only the investment margin in (4.5), not the consumption–saving budget set.

The solar ITC lowers the effective price of solar investment,

$$\tilde{p}^S_t = (1 - \phi^{ITC} \chi^S_t) p^S_t,$$

while wind and gas pay their full capital prices, $\tilde{p}_t^W = p_t^W$ and $\tilde{p}_t^G = p_t^G$. The wind PTC instead enters electricity-sector profits as a payment on eligible wind generation. Investment is subject to convex adjustment costs $\Psi^k(I_{jt}^k, K_{jt}^k)$, and wind and solar face rising regional deployment-cost curves $c_j^W(\cdot)$ and $c_j^S(\cdot)$ as a region builds out its most productive sites. Section 5.1 gives the functional forms and calibration.

Generation capital accumulates subject to irreversibility:

$$K_{j,t+1}^k = (1-\delta^k)K_{jt}^k + I_{jt}^k, \qquad I_{jt}^k \geq 0, \qquad k \in \{S, W, G\}. \tag{4.3}$$

To capture grandfathering of the wind PTC, I separately track subsidy-eligible wind capacity:

$$K_{j,t+1}^{W,PTC} = (1-\delta^W)(1-h^{exit})K_{jt}^{W,PTC} + I_{jt}^W \mathbf{1}[s_t = 1]. \tag{4.4}$$

Wind built while the subsidy is active enters both the physical and subsidy-eligible stocks; wind built during a lapse enters only the physical stock. The parameter h^{exit} governs the expiration of PTC eligibility.[8]

Let

$$\mathbf{K}_{jt} = (K_{jt}^S, K_{jt}^W, K_{jt}^{W,PTC}, K_{jt}^G).$$

The value of generation capital satisfies

$$\begin{aligned} V_t(\mathbf{K}_{jt}, s_t) = \max_{I^S, I^W, I^G \geq 0} \Big\{ & \Pi_{jt}^E(\mathbf{K}_{jt}, s_t) - \Xi_{jt}^S(I_{jt}^S) - \Xi_{jt}^W(I_{jt}^W) - \tilde{p}_t^G I_{jt}^G - \sum_k \Psi^k(I_{jt}^k, K_{jt}^k) \\ & + \mathbb{E}\left[\mathcal{M}_{t,t+1} V_{t+1}(\mathbf{K}_{j,t+1}, s_{t+1}) \mid s_t\right] \Big\}, \end{aligned} \tag{4.5}$$

subject to (4.3)–(4.4). Here $\Xi_{jt}^k(I^k) = \int_{K_{jt}^k}^{K_{jt}^k + I^k} c_j^k(x)\, dx$, $k \in \{S, W\}$, is solar and wind investment expenditure along the regional site-cost curves, and

$$\mathcal{M}_{t,t+1} = \beta \frac{\mathcal{U}_C(C_{j,t+1}, E_{j,t+1}^{HH})}{\mathcal{U}_C(C_{jt}, E_{jt}^{HH})}$$

is the household's stochastic discount factor.

The wind first-order condition illustrates how expectations about future subsidy avail-

[8]The model approximates the statutory ten-year PTC eligibility window with an annual exit rate for eligible capacity.

ability affect investment. For $I^W_{jt} > 0$,

$$\underbrace{c^W_j(K^W_{jt} + I^W_{jt}) + \Psi^W_I}_{\text{marginal cost}} = \underbrace{\mathbb{E}[\mathcal{M} q^W_{t+1} \mid s_t]}_{\text{energy value}} + \mathbf{1}[s_t = 1] \underbrace{\mathbb{E}[\mathcal{M} q^{W,PTC}_{t+1} \mid s_t]}_{\text{subsidy value}}. \tag{4.6}$$

When the subsidy is active, a marginal unit of wind earns both its electricity-market value and the value of the grandfathered subsidy stream.

Solar receives its subsidy as a reduction in up-front cost rather than as a payment on future output. For $I^S_{jt} > 0$,

$$\underbrace{c^S_j(K^S_{jt} + I^S_{jt}) + \Psi^S_I}_{\text{marginal cost, net of subsidy}} = \underbrace{\mathbb{E}[\mathcal{M} q^S_{t+1} \mid s_t]}_{\text{energy value}}. \tag{4.7}$$

Because the solar site-cost curve (5.1) is evaluated at the after-subsidy price $\tilde{p}^S_t = (1 - \phi^{ITC}\chi^S_t)p^S_t$, the credit lowers the cost of every solar site in proportion. The solar subsidy depends only on whether the subsidy is active in the year of construction. Expectations about future subsidy availability affect solar investment through the timing incentive (a project built in a lapse forgoes a credit that may return) and through future electricity prices in q^S_{t+1}, but not through a subsidy stream attached to installed capacity.

4.3 Production firms and the national capital market

A representative firm in region j produces the final good from capital, labor, and electricity. It combines a Cobb–Douglas capital–labor bundle with electricity in a constant-returns CES technology,

$$Y_{jt} = Z_{jt}\left[(1-\gamma_E)^{1/\sigma}\left(K^{\hat{\alpha}_K}_{P,jt} L^{\hat{\alpha}_L}_{jt}\right)^{\frac{\sigma-1}{\sigma}} + \gamma_E^{1/\sigma}\left(E^{firm}_{jt}\right)^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}, \tag{4.8}$$

where

$$\hat{\alpha}_K = \frac{\alpha_K}{\alpha_K + \alpha_L}, \qquad \hat{\alpha}_L = \frac{\alpha_L}{\alpha_K + \alpha_L}.$$

Production capital is supplied through a national market. Households hold financial wealth A_{jt} and firms rent production capital $K_{P,jt}$ at the common rental rate r^K_t, which

clears the national capital market:

$$\sum_j K_{P,jt} = \sum_j A_{jt}. \tag{4.9}$$

4.4 The electricity sector

The household-generator owns generation capacity in hydro, nuclear, wind, solar, gas, and coal. It chooses investment only in wind, solar, and gas. Hydro and nuclear are fixed stocks, while coal follows an exogenous retirement path and no new coal is built.

Dispatch. Generation *capacity* is distinct from electricity *generation*. Figure 6 illustrates the merit-order dispatch curve.

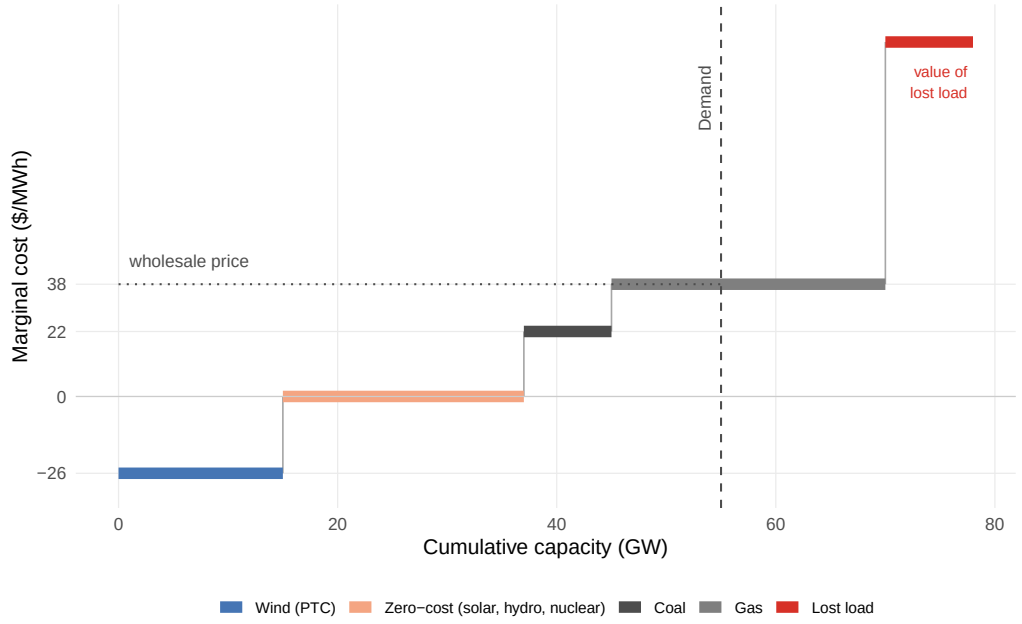


Figure 6: The merit-order dispatch curve.

Notes. Merit-order supply curve for a region in a single period. Illustrative capacities and costs.

Within-period electricity demand follows a Pareto distribution with region-specific tail index θ_j and scale $x_{jt}^{\min}$,

$$\Pr(D_{jt} > d) = \left(\frac{x_{jt}^{\min}}{d}\right)^{\theta_j}, \qquad \mathbb{E}[D_{jt}] = \frac{\theta_j}{\theta_j - 1} x_{jt}^{\min}. \tag{4.10}$$

A lower θ_j implies a heavier upper tail of demand, so peak load is larger relative to base load. Annual household and firm electricity demand is price elastic,

$$E_{jt}^{firm} = \bar{\alpha}_E Y_{jt}(p_{jt}^E)^{-\sigma}, \qquad E_{jt}^{HH} = \bar{\alpha}_h C_{jt}(p_{jt}^E)^{-\sigma}, \tag{4.11}$$

and $x_{jt}^{\min}$ is chosen so that expected hourly demand aggregates to annual electricity demand.

The expected wholesale price is the expected marginal cost of serving load:

$$\begin{aligned} \bar{p}_{jt}^E = {} & mc_{jt}^{lo} \Pr(D_{jt} > Q_{jt}^0) \\ & + (mc_{jt}^{hi} - mc_{jt}^{lo}) \Pr(D_{jt} > Q_{jt}^0 + K_{jt}^{lo}) \\ & + (V_j - mc_{jt}^{hi}) \Pr(D_{jt} > \bar{Q}_{jt}), \end{aligned} \tag{4.12}$$

where Q_{jt}^0 is zero-marginal-cost capacity, mc_{jt}^{lo} and mc_{jt}^{hi} are the marginal costs of the cheaper and more expensive fossil technologies, and $\bar{Q}_{jt}$ is total capacity. Additional renewable capacity expands the zero-cost portion of the stack, makes fossil technologies marginal less often, and therefore lowers the electricity price earned by generation. Appendix A gives the corresponding closed-form expression under Pareto demand.

Capacity adequacy. I model a capacity-adequacy constraint which requires each region to maintain enough reliable generation capacity to meet forecast demand at a given loss-of-load-probability target. Because gas is the only dispatchable technology that can be built endogenously, it fills any capacity shortfall when existing baseline generation capacity is insufficient. The multiplier on the constraint is a capacity price p_{jt}^{cap}, which captures the value of additional reliable capacity. The requirement is based on each region's exogenous forecast demand. Appendix A.3 defines the constraint and capacity price formally.

Trade. Regions trade electricity within, but not across, the three interconnections. Let $X_{jt} = G_{jt} - \bar{E}_{jt}$ denote region j's net exports. The local electricity price is the interconnection hub price net of a congestion basis,

$$p_{jt}^E = \bar{p}_{mt}^E - \gamma \frac{X_{jt}}{\overline{TL}_j}, \tag{4.13}$$

where $\overline{TL}_j$ measures transmission capacity. An exporter's local price falls relative to the hub price as congestion increases, while an importer pays a higher local price. Each interconnection's hub price clears aggregate net trade among its member regions. Transmission costs

are real resource costs; congestion-basis payments are transfers and are rebated lump-sum through $\mathcal{R}_{jt}$. Appendix A gives the complete electricity-sector profit and trade accounting.

National electricity-sector CO_2 emissions are

$$E_t = \sum_j \left(\xi^G gen^G_{jt} + \xi^C gen^C_{jt}\right). \tag{4.14}$$

4.5 Equilibrium

Given the exogenous paths and policy transition matrix Π, a recursive competitive equilibrium is a sequence of household allocations, generation and production capital, investment policies, electricity generation and trade, and prices such that: *(i)* households optimize consumption, saving, and generation investment; *(ii)* final-good firms maximize profits; *(iii)* generation is dispatched in merit order and electricity markets clear within each interconnection; *(iv)* the national capital market and regional labor markets clear; *(v)* the federal budget and aggregate resource constraint hold; and *(vi)* expectations are consistent with the policy transition matrix Π. The economy converges to a terminal stochastic steady state. Appendix A states the equilibrium conditions formally, and Appendix C describes the computational solution.

4.6 Welfare

I measure welfare as the present value of real income, household spending on the final good and on electricity, net of the environmental damages from power-sector emissions. The environmental damages have two components: global climate damages from CO_2 and local air-pollution mortality damages from fossil generation,

$$\begin{aligned} W^C &= \sum_t \beta^{t-2010} \sum_j \left(C_{jt} + p^E_{jt} E^{HH}_{jt}\right), \qquad W^{\mathrm{CO_2}} = \sum_t \left(\frac{1}{1+r_c}\right)^{t-2010} \mathrm{SCC}_t \, E_t, \\ W^{\mathrm{AP}} &= \sum_t \left(\frac{1}{1+r}\right)^{t-2010} \sum_j \left(d_{jC}\, gen^C_{jt} + d_{jG}\, gen^G_{jt}\right), \end{aligned} \tag{4.15}$$

and total welfare is $W^{\$} = W^C - W^{\mathrm{CO_2}} - W^{\mathrm{AP}}$. Real income values household electricity at its market price and is discounted at the household rate β. It is a money-metric approximation to the utility-based welfare in (4.1); I also report the corresponding consumption-equivalent

change. The climate component values each year's CO_2 emissions E_t at the EPA social cost of carbon, discounted at 2% (U.S. Environmental Protection Agency, 2023). The local air-pollution component values regional coal and gas generation gen_{jt}^C and gen_{jt}^G at regional fuel-specific marginal damage coefficients d_{jC} and d_{jG}, and discounts the resulting contemporaneous mortality damages at the private rate r. The marginal damage coefficients come from the AP4 integrated assessment model and value $PM_{2.5}$-related mortality; Appendix G.3 gives their construction.

Subsidies, the taxes that finance them, and congestion rebates are transfers and therefore cancel from aggregate welfare. Generation investment, adjustment costs, fossil fuel use, transmission, and production-capital accumulation are real resource costs. Appendix A.5 gives the equivalent aggregate resource constraint.

5 Calibration

I calibrate the model in three steps. First, I set technology, demand, emissions, and regional electricity-market parameters using external data. Second, I calibrate the investment and site-cost parameters to reproduce national U.S. wind and solar capacity levels over 2010–2024. Third, I calibrate firms' expectations about future subsidy availability using the historical policy record and transitions in federal political control. Tables 2 and 3 summarize the data sources and parameter values.

5.1 External parameters and exogenous paths

The exogenous paths come from standard energy datasets. Wind, solar, and gas capital costs follow the NREL Annual Technology Baseline (Moderate) projection, joined to historical overnight costs from LBNL and EIA. Electricity demand follows the EIA Annual Energy Outlook 2025 reference case. Fuel costs and the exogenous coal-retirement path come from EIA. Appendix Figure 12 plots these paths.

Regional renewable productivity is measured using realized capacity factors, calculated as net generation from EIA-923 divided by nameplate capacity from EIA-860 and averaged over 2019–2024. I use NREL developable-resource data to calibrate the regional scale of wind and solar site availability. I use EIA-930 to calculate the shape of within-year electricity demand

and calibrate the parameters for interregional trade. Appendix F describes the construction of each input and Table 11 reports regional summary statistics.

The remaining externally calibrated parameters come from engineering estimates and the empirical literature. Technology-specific depreciation rates and renewable capacity credits come from NREL, emissions factors from EPA eGRID, and production cost shares from BEA. The wind PTC and solar ITC are set to their legislation values. Panel A of Table 3 reports the full set of externally calibrated parameters.

Table 2: Data sources.

Dataset	Source	Years	Use in the model
Existing capacity	EIA-860	2010–2024	Initial stocks $K^k_{j,0}$, coal-retirement path
Generation & fuel	EIA-923	2010–2024	Capacity factors CF^k_j; fuel marginal costs
Hourly demand & interchange	EIA-930	2021–2023	Pareto demand shape θ_j; trade capacity $\overline{TL}_j$
Electricity demand path	EIA AEO 2025 Ref.	2024–2050	Demand growth (electrification growth)
Capital-cost projection	NREL ATB 2024 (Moderate)	2025–2050	Renewable and gas cost decline
Developable resource	NREL land-based resource	2023	Site-cost scales $\bar{K}^k_j$
Wholesale prices	ICE, NYISO	2021–2025	Interconnection trade slope s_j
Emissions factors	EPA eGRID	2022	ξ_G, ξ_C
Household counts	U.S. Census	2020	Regional L_j; state-to-region aggregation
Policy history	PTC/ITC statutes, IRS notices	1992–2025	Eligibility; forward Markov chain
Region geography	EIA / Census TIGER	2019	13-region map
Local air-pollution damages	AP4 / EIA-923	2017	$PM_{2.5}$-related mortality damage coefficients

Notes. The AEO 2025 demand path and the NREL ATB 2024 cost projection are published through 2050 and held (or extrapolated along their terminal trend) over the remaining model horizon.

5.2 Internal calibration

Four parameters govern how quickly renewable investment responds to changes in profitability. Investment adjustment costs are quadratic in the investment rate,

$$\Psi^k(I, K) = \frac{\psi^k}{2}\left(\frac{I}{K}\right)^2 K,$$

Table 3: Model parameters.

Parameter	Symbol	Value
Panel A. Externally calibrated (data, engineering, tax code)		
Discount factor	β	0.9615 ($\rho \approx 4\%$)
Wind, solar depreciation	δ^W, δ^S	0.04, 0.033 (NREL ATB)
Gas, production-capital depreciation	δ^G, δ_P	0.03, 0.05
PTC value	ϕ^{PTC}	\$26/MWh (§45)
ITC rate	ϕ^{ITC}	30% (§48)
Solar capital cost path	p_t^S	\$4.27M→\$0.79M/MW
Wind capital cost path	p_t^W	\$2.46M→\$1.10M/MW
Gas capital cost path	p_t^G	\$1.33M→\$0.74M/MW
Elasticity	σ	0.3
Emissions factors	ξ^G, ξ^C	0.40, 0.95 tCO_2/MWh (eGRID)
Social cost of carbon	SCC	\$213/tCO_2 (EPA 2023)
PTC vintage duration	h^{exit}	0.1 (§45)
Site-cost curvature	η	2
Capacity factors	CF_j^k	wind 0.04–0.38, solar 0.19–0.32
Trade slope (congestion)	$s_j = \gamma/\overline{TL}_j$	0.24–8.6, $\gamma \approx 2$ (ICE/NYISO, EIA-930)
Reliability target	ε	0.002 (NERC)
Capacity credit	ζ^W, ζ^S	0.15, 0.10 (NREL)
Cost shares (capital/labor/electricity)	$\alpha_K, \alpha_L, \alpha_E$	0.33, 0.65, 0.02 (BEA)
Electricity demand weights	$\bar{\alpha}_h, \bar{\alpha}_E$	$0.0014 = 0.02 \times 45^{-0.7}$ (2% share)
Panel B. Internally calibrated to the 2010–2024 capacity growth		
Wind, solar adjustment cost	ψ^W, ψ^S	2×10^5 each
Gas adjustment cost	ψ^G	10^7
Wind site-cost scale	κ^W	0.08 (2024 wind ≈143 GW)
Solar site-cost scale	κ^S	0.22 (2024 solar ≈136 GW)

Notes. Panel A is set from external data, engineering studies, and the tax code. Panel B's four deployment-cost parameters are calibrated to the observed 2010–2024 renewable capacity growth, matching the 2024 capacity levels. The policy Markov chain is calibrated separately (Section 5.3).

and capture the increasing cost of expanding generation capacity rapidly. Wind and solar deployment costs also rise as a region exhausts its most productive sites:

$$c_j^k(K_{jt}^k) = \tilde{p}_t^k \left[1 + \left(\frac{K_{jt}^k}{\bar{K}_j^k}\right)^{\eta}\right], \qquad k \in \{S, W\}, \tag{5.1}$$

where $\bar{K}_j^k$ is proportional to region j's developable wind or solar resource and I set $\eta = 2$.

I calibrate the wind and solar adjustment-cost parameters and the wind and solar site-cost scales jointly to reproduce national wind and solar capacity levels over 2010–2024. Adjustment costs determine how quickly investment can respond when profitability changes.

The site-cost scales determine how rapidly increasingly expensive locations are reached as cumulative deployment rises.

5.3 Policy expectations

The availability of the subsidies follows a two-state Markov process,

$$s_t \in \{0, 1\},$$

with π_{11} denoting the probability that an active subsidy remains active and π_{00} the probability that a lapse continues for another year.

I use the history of federal political control to discipline π_{00}. I classify each Congress since 1993 as a Democratic trifecta, a Republican trifecta, or divided government and estimate transitions among the three states (Table 10). I map a Democratic trifecta into the active subsidy state and a Republican trifecta into the lapsed state; divided government leaves the current policy unchanged.

I convert the estimated political process to annual transition probabilities by preserving its expected spell lengths. This gives $\pi_{11} = 0.90$ (an expected ten years active) and $\pi_{00} = 0.94$ (an expected seventeen-year lapse). The robustness analysis also considers faster renewal and permanent repeal. Section 7 shows how the results change across this range.

5.4 Validation: untargeted moments

The calibration uses national wind and solar capacity levels from 2010–2024, but does not target emissions, generation shares, electricity prices, the location of renewable capacity, or electricity trade.

At the national level, the model produces 1.52 Gt of power-sector CO_2 emissions in 2024, compared with 1.5 Gt in the data. Wind supplies 10% of generation in both the model and the data, hydro and nuclear supply 25% in the model and 24% in the data. Fossil fuel together supply 57% and 58%, respectively. The main difference is solar: its share is 8% in the model and 5% in the data. The model's average wholesale electricity price also falls within the observed 2021–2024 range.

Table 4: Untargeted validation: model versus data.

Moment	Model	Data
Panel A. Aggregate moments		
Power-sector CO_2, 2010 (Gt/yr)	2.15	2.2
Power-sector CO_2, 2024 (Gt/yr)	1.52	1.5
Wholesale price, 2021–2024 ($/MWh)	41	35–45
Wind generation share, 2024 (%)	10	10
Solar generation share, 2024 (%)	8	5
Baseload generation share, 2024 (%)	25	24
Fossil generation share, 2024 (%)	57	58
Panel B. Spatial agreement		
Wind capacity across regions, Spearman ρ	0.96	
Solar capacity across regions, Spearman ρ	0.57	
Net-trade sign	8/12 regions	

Notes. Regional capacity statistics report the Spearman rank correlation between modeled and observed 2024 capacity across the 13 regions. The trade statistic reports the number of regions for which the model reproduces the observed sign of net trade; Texas is excluded because ERCOT has no interconnection trade in the model.

The regional allocation of renewable capacity is not a calibration target. However, the Spearman rank correlation between modeled and observed wind capacity across the 13 regions in 2024 is $\rho = 0.96$. The match is weaker for solar, with $\rho = 0.57$. The model gets the direction of net electricity trade right in 8 of the 12 regions that trade across regional boundaries (excluding Texas as it does not trade).

Figures 14 and 15 provide the corresponding dynamic and regional comparisons.

6 Counterfactuals

All counterfactuals share the realized 2010–2027 policy history and therefore enter 2028 with identical state variables.[9] The policy experiments branch in 2028, when the OBBBA repeal takes effect.

I consider three main policy regimes: a stable subsidy, an unstable subsidy, and no subsidy.

[9]The model transition begins in 2010 and is calibrated and validated against the observed electricity transition. The 2010 start date does not affect differences across the post-2028 policy experiments.

6.1 Policy regimes

Stable subsidy. The subsidy remains available in every period, $s_t = 1$. Wind receives the production tax credit of \$26/MWh and solar receives the 30% investment tax credit. Firms know that the subsidy will remain available when making investment decisions. The present value of government spending under this policy is \$231 billion.

Unstable subsidy. The subsidy availability follows the two-state Markov process calibrated in Section 5.3. The transition probabilities are known but the dates of future lapses and renewals are not. I simulate 500 subsidy paths from this process and report outcomes averaged across them. I keep the solar ITC at 30% and set the active-state wind PTC to \$36.4/MWh, so that expected present-value subsidy spending under the unstable subsidy also equals \$231 billion.

No subsidy. The subsidy is permanently unavailable, $s_t = 0$, and firms know that it will not return.

6.2 Counterfactual comparisons

My main welfare comparison is

$$W^{\text{perm}} - W^{\text{unc}},$$

which I call the *value of a stable subsidy*. The stable and unstable subsidies have the same expected present-value subsidy spending.

I also report

$$W^{\text{unc}} - W^{\text{none}},$$

the value of the unstable subsidy relative to no federal subsidy. The two differences add up to the value of the stable subsidy relative to no subsidy:

$$W^{\text{perm}} - W^{\text{none}} = (W^{\text{perm}} - W^{\text{unc}}) + (W^{\text{unc}} - W^{\text{none}}).$$

6.3 Perfect-foresight counterfactuals

To separate uncertainty about future policy dates from the effects of changes in subsidy availability themselves, I construct a full-foresight counterfactual. I draw subsidy paths from the same Markov process that governs the unstable subsidy in the baseline. For each realized path, I reveal the complete future subsidy schedule to firms in 2028 and solve the model under perfect foresight. Averaging outcomes across these paths gives expected welfare under the same distribution of future subsidy realizations as the baseline, but without uncertainty about when policy changes occur.

This experiment decomposes the value of a stable subsidy into two components. Let W^{perm} denote welfare under the stable subsidy, W^{unc} welfare under the unstable subsidy, and $\mathbb{E}_\omega[W^{\text{PF}}(\omega)]$ expected welfare when each realized policy path ω is known in advance. Then

$$W^{\text{perm}} - W^{\text{unc}} = \left(W^{\text{perm}} - \mathbb{E}_\omega[W^{\text{PF}}(\omega)]\right) + \left(\mathbb{E}_\omega[W^{\text{PF}}(\omega)] - W^{\text{unc}}\right).$$

The first term captures the value of a stable subsidy when future policy dates are known, while the second captures the value of resolving uncertainty about the future policy path.

To illustrate how firms respond to a known schedule, I also solve the model along one schedule: the subsidy is off in 2028–2033, on in 2034–2043, and off from 2044 on. I adjust the active-state wind PTC so that present-value subsidy spending on this schedule ($232.0 billion) is close to spending under the stable subsidy ($230.9 billion).

7 Results

7.1 The value of a budget-neutral stable subsidy

At the same expected budget, the stable subsidy produces more renewable capacity and less gas than the unstable subsidy. By 2070, the stable subsidy yields 137 GW more renewable capacity and 37 GW less gas (Figure 7). The lower renewable capacity under the unstable subsidy translates into higher fossil generation and emissions (Figure 8). Cumulatively, the unstable subsidy produces 2.29 Gt more CO_2 through 2070, or 3.2% relative to the stable subsidy, equivalent to roughly a year and a half of U.S. power-sector emissions in 2024.

Most of the emissions difference does not come from additional gas capacity. Existing

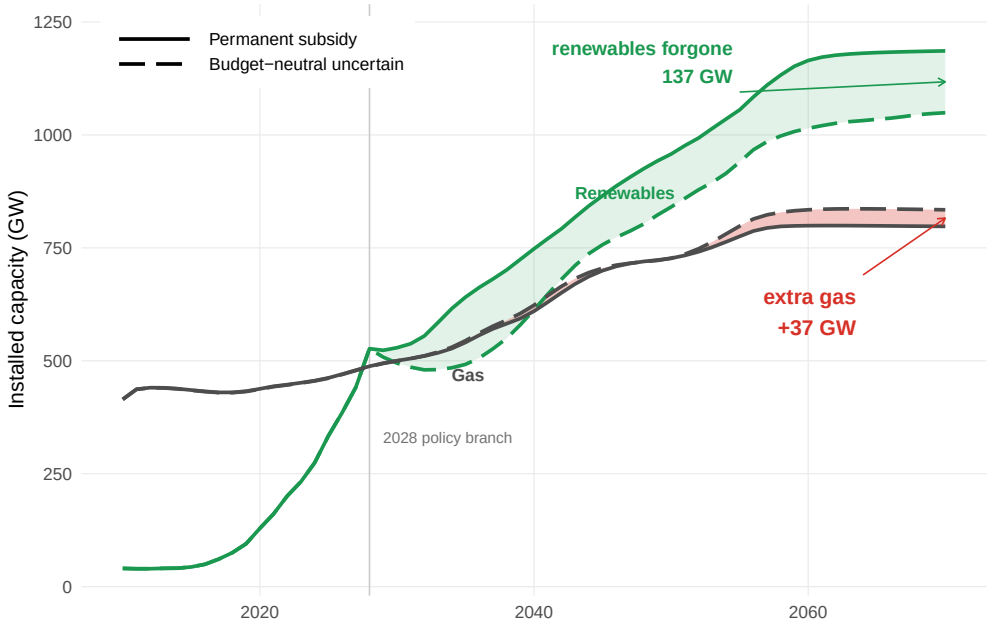


Figure 7: **Renewable and gas capacity under the stable subsidy and the budget-neutral unstable subsidy.**

Notes. Installed renewable (wind plus solar) and gas capacity under the stable subsidy (solid) and the budget-neutral unstable subsidy (dashed).

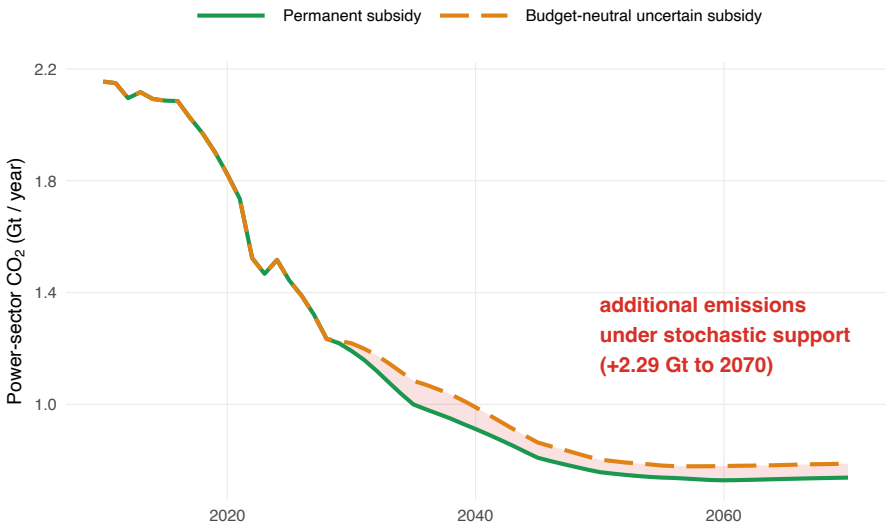


Figure 8: Annual power-sector CO_2 by policy regime.

Notes. Annual U.S. power-sector CO_2 emissions under the stable subsidy and the budget-neutral unstable subsidy.

fossil plants generate more when renewable investment is delayed. Additional generation from existing coal and gas accounts for 1.83 Gt, or 80% of the cumulative emissions gap. By fuel, 52% of the total gap comes from gas and 48% from coal. The remaining 0.46 Gt (20%)

comes from additional gas capacity built under the unstable subsidy.

Table 5: The value of budget-neutral continuous subsidy, decomposed.

($B, present value)	Permanent − uncertain	Uncertain − no subsidy
Environmental gains	**406**	**227**
Climate damages	360	203
existing coal	177	95
existing gas	116	63
new gas	66	45
Local air-pollution mortality	46	24
coal	39	20
gas	7	4
Consumption (net change)	**−113**	**−134**
Renewable & baseload income	144	66
Wages	16	12
Gas & coal profit	−23	−9
Generation investment	−251	−203
Value	**292**	**93**
Value excluding local air pollution	246	69

Notes. Present value in billions of real 2022 dollars, horizon through 2070. Consumption is the net change in the present value of consumption; income, wages, and profit raise it, while generation investment enters negatively. Value = environmental gains + net change in consumption. Column 1 is the value of continuous support (versus a budget-neutral uncertain subsidy); column 2 is the value of the uncertain subsidy versus no subsidy.

Table 5 decomposes the welfare change. Replacing the unstable subsidy with a budget-neutral stable subsidy generates $292 billion in net benefits. The stable subsidy avoids $406 billion in environmental damages: $360 billion in climate damages from CO_2 and $46 billion in mortality damages from local air pollution. Most of the climate benefit comes from lower generation by existing coal and gas, with the remainder from gas capacity that is never built.

These benefits are partly offset by a consumption cost. The stable subsidy raises renewable investment and lowers consumption-equivalent welfare by 0.12%, a present-value consumption cost of $113 billion (Table 5). Excluding local air-pollution damages, the value of a stable subsidy is $246 billion; valuing only avoided CO_2, the stable subsidy breaks even at a social cost of carbon of $\$67/tCO_2$.

Including avoided local air-pollution mortality raises the value of the unstable subsidy relative to no subsidy from $69 billion to $93 billion. The value of a stable subsidy remains positive under faster subsidy renewal, permanent repeal, alternative reporting horizons, and

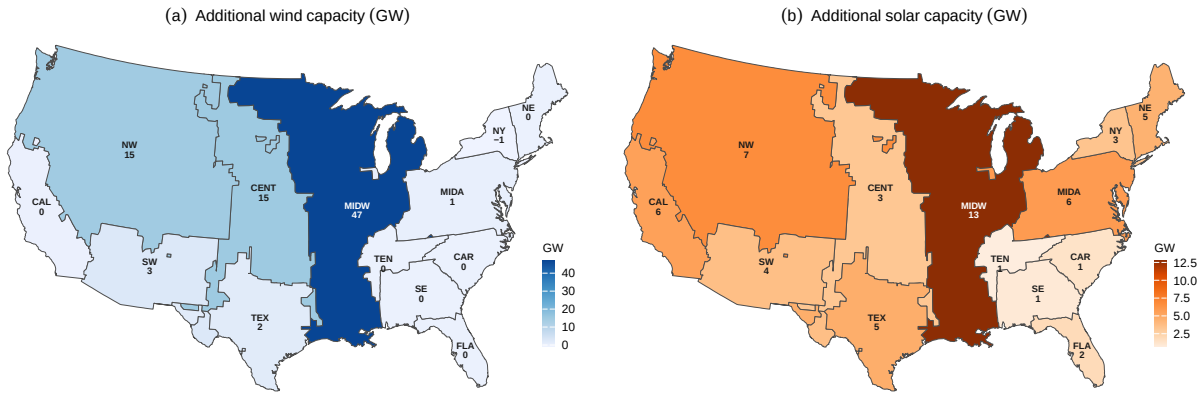


Figure 9: Regional capacity response to the stable subsidy.

Notes. Additional installed capacity under the stable subsidy relative to the budget-neutral unstable subsidy in 2070, by EIA region. Panel (a) reports wind and panel (b) solar.

discounting climate damages at the market rate, ranging from \$56 billion to \$385 billion (\$48–\$315 billion excluding local air pollution); Appendix H reports these exercises.

7.2 Regional heterogeneity

The aggregate investment response is unevenly distributed across regions (Figure 9). The wind response is concentrated in the interior: the Midwest, Central, and Northwest together build 77 GW more wind under the stable subsidy. In contrast, regions with poor wind resources, such as Tennessee and Florida, build little additional wind. The solar response is more geographically dispersed. The stable subsidy increases solar capacity across nearly every region, with the largest increase in the Midwest and substantial responses in the Northwest, California, the Mid-Atlantic, New England, and Texas. The Midwest has the largest combined wind and solar capacity response: about 60 GW of additional renewable capacity by 2070 (47.1 GW of wind and 13 GW of solar).

The effect of the stable subsidy is largest neither in regions with the best renewable resources nor in those with the worst. Where renewable projects are profitable without the federal subsidy, investment can continue during a lapse. At the other extreme, where renewable resources are poor, even the subsidy may not be sufficient to make additional projects profitable. The stable subsidy matters most for the marginal regions, where the subsidy determines whether additional renewable capacity is built or not.

The emissions response also depends on the generation capacity that was already in

place when the policies diverge in 2028. When renewable investment is delayed, regions with substantial existing fossil capacity can meet more electricity demand by running those plants more intensively. The Midwest has the largest response due to large initial carbon-intensive fossil capacity (especially coal) which generates additional emissions when the renewable investment is delayed.

The environmental consequences depend not only on how much fossil generation responds, but also on which fossil technologies respond. This distinction is particularly important for local air pollution. Using plant-specific marginal damages from AP4, the generation-weighted $PM_{2.5}$-related mortality damage of coal generation is approximately \$105/MWh nationally, compared with \$11/MWh for gas (in 2020 dollars). Regions in which the stable subsidy reduces coal generation therefore account for a disproportionate share of the local air-pollution benefit.

Electricity trade creates an additional source of spatial spillovers: renewable investment in one region can reduce fossil generation elsewhere. Regions trade within the Eastern and Western interconnections, while Texas forms a separate interconnection. California illustrates this mechanism: it is a net importer, and its own change in renewable-capacity is relatively small. Under the stable subsidy, however, the Northwest and Southwest (which both have a high solar suitability) build more renewable capacity and supply more clean electricity within the Western Interconnection. California imports part of this additional generation and runs its own gas capacity less. As a result, California gains \$38 billion from the stable subsidy even though it adds only about 6 GW of renewable capacity itself. In contrast, Central and the Southwest lose slightly (\$12 billion and \$2 billion) because they are net exporters of clean electricity and bear part of the additional investment cost (Table 6).

Table 6 reports the corresponding regional decomposition. I assign the climate component according to the region in which power-sector CO_2 emissions change. Because CO_2 damages are global, this allocation is an accounting decomposition of the national climate benefit rather than a measure of local environmental benefits or regional welfare incidence. Avoided local air-pollution mortality damages are attributed to the source region of the emitting plants; Appendix G.3 describes their construction.

The regional decomposition confirms the importance of the existing fossil fleet. The Midwest gains \$154 billion from the stable subsidy, the most of any region, and most of this gain comes from avoided climate and local air-pollution damages as coal generation falls. Of the Midwest's \$129 billion in climate gains under the stable subsidy, \$98 billion comes

Table 6: The value of continuous support by region, decomposed (continuous support versus budget-neutral uncertain subsidy).

	Climate gains			Local poll.†	Consumption					Total
Region	Coal	Gas	Total	Avoided	Invest.	Income	Wages	Profit	Total	value
Midwest	98	30	129	25	−59	66	−3	−4	0	154
Texas	38	8	45	12	−76	72	−8	1	−12	46
California	0	54	54	1	−24	10	4	−7	−17	38
Mid-Atlantic	0	29	29	2	−16	7	0	0	−9	23
Tennessee	0	4	4	0	7	1	0	0	8	12
New England	0	17	17	0	−12	6	0	−1	−7	11
Florida	1	14	14	1	−8	3	0	0	−5	10
New York	0	11	11	0	−8	4	1	−2	−5	6
Carolinas	6	0	7	1	−3	0	1	0	−2	5
Southeast	1	3	4	0	−2	1	0	0	−2	2
Northwest	20	3	24	1	−19	−13	13	−7	−26	0
Southwest	8	3	11	1	−13	−3	4	−2	−14	−2
Central	5	5	10	1	−17	−9	4	0	−23	−12
National	**177**	**183**	**360**	**46**	**−251**	**144**	**16**	**−23**	**−113**	**292**

Notes. Present value in billions of real 2022 dollars, by region, sorted by total value. Climate gains value the region's avoided CO_2 at the EPA year-specific social cost of carbon discounted at 2%, split into existing coal and total gas. Consumption is the net change in the present value of the region's consumption; income, wages, and profit raise it, while generation investment enters negatively. Total value = climate gains + local air pollution + consumption; every column sums to the National row, which matches Table 5. †Avoided local air-pollution ($PM_{2.5}$) mortality damages attributed to emissions from fossil plants *located in* the region (AP4, discounted at 4%). Because pollution disperses across regions, much of the avoided mortality occurs downwind, so these are not health benefits received by residents of the source region.

from reduced generation by existing coal and another $17 billion from existing gas; only $13 billion comes from avoided new gas. The concentration of the response in coal also makes the Midwest particularly important for local air pollution. Plants located in the Midwest account for approximately $25 billion, or 54%, of national avoided $PM_{2.5}$-related mortality damages (Appendix G.3). These damages are attributed to the location of the emitting plants and do not measure health benefits received specifically by Midwest residents.

7.3 Policy switching versus uncertainty about timing

The stable and unstable subsidies differ in two respects: under the unstable subsidy, availability changes over time, and firms do not know when the changes will occur. The perfect-foresight counterfactual keeps the switching but removes the uncertainty about its timing (Table 7). The value of a stable subsidy therefore splits into the effect of switching, $W^{\text{perm}} - \mathbb{E}_\omega[W^{\text{PF}}(\omega)]$, and the effect of uncertainty about timing, $\mathbb{E}_\omega[W^{\text{PF}}(\omega)] - W^{\text{unc}}$.

Table 7: Policy regimes in the perfect-foresight decomposition.

Subsidy availability	Timing known	Timing uncertain
Always available	Stable, W^{perm}	—
Changes over time	Switching, perfect foresight, $\mathbb{E}_\omega[W^{\text{PF}}(\omega)]$	Unstable, W^{unc}

Notes. Under perfect foresight, the subsidy follows the same schedules as under the unstable subsidy, drawn from the same Markov process, but firms learn the complete schedule in 2028. $\mathbb{E}_\omega$ averages welfare over the drawn schedules ω.

Table 8 reports the decomposition. Knowing the policy dates does not eliminate the value of a stable subsidy: relative to the switching subsidy under perfect foresight, the stable subsidy is worth $355 billion. Foresight changes the timing of investment rather than the amount of abatement: avoided climate damages are nearly the same with and without foresight ($353 billion versus $360 billion). The difference is in resource costs. When firms know the dates, they concentrate more investment in the years before a known deadline, and convex adjustment costs make this bunching expensive: consumption is $74 billion lower under perfect foresight than under the unstable subsidy. Resolving uncertainty about timing therefore lowers welfare by $63 billion. Most of the value of a stable subsidy comes from removing policy switching itself, and uncertainty about timing partially offsets the cost of switching.

Figure 10 shows where the additional resource cost comes from, using one schedule drawn from the policy process: the subsidy is off in 2028–2033, on in 2034–2043, and off from 2044

Table 8: Decomposition of the value of a stable subsidy.

($B)	Climate	$PM_{2.5}$	Consumption	Value
Value of a stable subsidy	360	46	−113	292
= Effect of switching	353	41	−39	355
+ Effect of uncertainty about timing	7	5	−74	−63

Notes. Present value in billions of real 2022 dollars, horizon through 2070. The value of a stable subsidy is $W^{\text{perm}} - W^{\text{unc}}$, the effect of switching is $W^{\text{perm}} - \mathbb{E}_\omega[W^{\text{PF}}(\omega)]$, and the effect of uncertainty about timing is $\mathbb{E}_\omega[W^{\text{PF}}(\omega)] - W^{\text{unc}}$. Climate and $PM_{2.5}$ are avoided climate damages and avoided $PM_{2.5}$-related mortality damages; consumption is the net change in the present value of consumption. Value = climate + $PM_{2.5}$ + consumption.

on. Both policies share the realized history through 2027, including the investment rush before the OBBBA deadline. Under the stable subsidy, renewable investment rises from about 15 GW in 2028 to about 50 GW by 2032 and then stays between 45 and 70 GW per year. With the known schedule, firms build almost nothing during the 2028–2033 lapse, because they know the subsidy returns in 2034. Investment jumps to about 90 GW per year when the subsidy returns and peaks at about 170 GW in 2043, the last year in which new wind qualifies for the production tax credit. From 2044 on, investment falls to almost zero and recovers only gradually, to a lower level than under the stable subsidy. Most of the 2043 peak is wind (Appendix H.6).

7.4 Discussion

The model abstracts from several mechanisms and margins that could affect the magnitude, timing, and spatial distribution of the response to changes in federal renewable subsidies. I discuss the main ones here.

Coal retirements. Coal retirements are exogenous in the model. In practice, utilities can postpone planned retirements when additional generation is needed. During a subsidy lapse, delayed renewable investment could therefore lead utilities to keep existing coal plants operating for longer. Allowing endogenous retirement could increase reliance on existing coal generation while renewable investment is low and reduce the need for additional gas capacity. Because coal is more emissions-intensive than gas, the effect on cumulative emissions is ambiguous even if less new gas capacity is built.

Gas construction lags. New gas capacity can be built within one period in the model, while in practice gas projects face permitting and construction lags. Longer construction

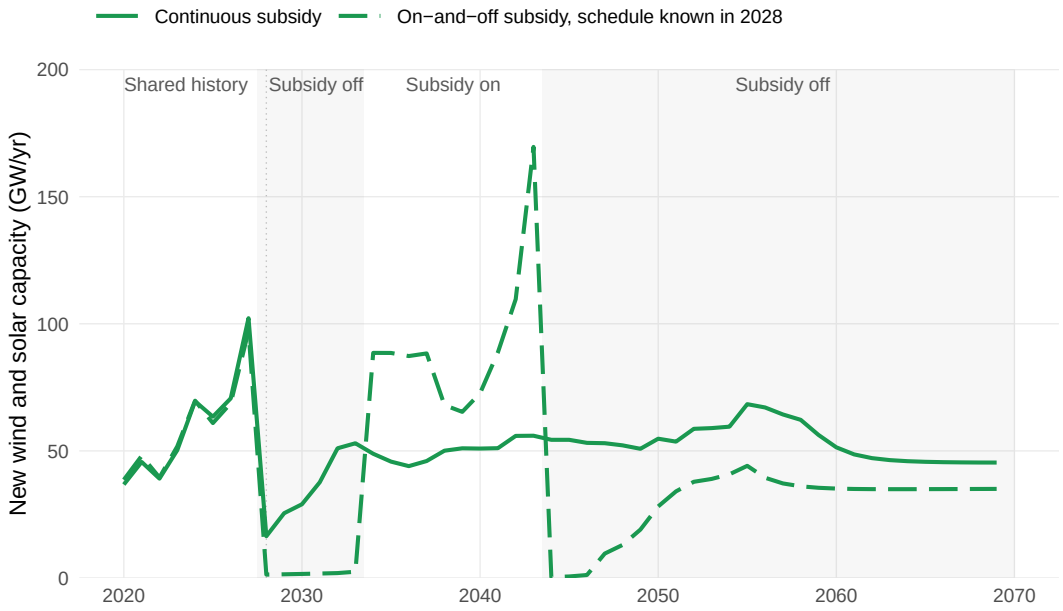


Figure 10: Renewable investment under the stable subsidy and a known switching schedule.

Notes. Annual wind and solar investment (GW per year). Both policies share the realized history through 2027. Under the switching subsidy with perfect foresight, firms learn the complete future subsidy schedule in 2028: the subsidy is off in 2028–2033, on in 2034–2043, and off from 2044 on. Shaded areas mark the years in which the subsidy is off under this schedule; under the stable subsidy it is always on. This is a transition path from one particular draw; the decomposition in Table 8 averages over all schedules drawn from the policy process.

times would limit the ability to build gas capacity in response to delayed renewable investment and shift more of the short-run response toward greater utilization of the existing fossil fleet. They could also reduce the persistence of the capital-stock difference, because less additional gas capacity would be built during a subsidy lapse. In the baseline counterfactual, 80% of the cumulative emissions difference already comes from greater utilization of existing coal and gas rather than additional gas capacity.

Learning-by-doing. The model does not include learning-by-doing. If additional U.S. deployment lowers subsequent renewable costs, maintaining the subsidy could add more current investment and lower future investment costs, amplifying the difference in deployment across policy paths. The magnitude of this channel depends on how strongly technology costs respond to U.S. deployment rather than global manufacturing and deployment.

Storage. The model has no grid storage. Storage complements renewables by shifting electricity from low-price to high-price hours. Including it could affect the stable-subsidy result in two directions. In some regions, storage could make renewable projects profitable without the federal subsidy, allowing investment to continue during a lapse and narrowing

the gap between the stable-subsidy and unstable-subsidy paths. In regions where investment would still stop during a lapse, however, the forgone renewable capacity would be more valuable, widening the gap. The net effect is therefore ambiguous.

Intraday renewable generation. Wind and solar generation are represented using annual capacity factors. The model therefore does not capture the covariance between renewable output and electricity demand or differences in the timing of wind and solar generation. An hourly representation could raise the value of solar where output coincides with peak demand, but lower it as midday generation becomes abundant and wholesale prices fall. These effects would also interact with storage. Incorporating hourly renewable profiles would refine the technology mix and dispatch response, although the policy mechanism studied here would not be affected.

Nuclear generation. Nuclear capacity is fixed in the model. Allowing new nuclear investment would provide another source of firm low-carbon capacity and could reduce the gas investment induced by slower renewable deployment.

Electricity-market institutions. Each region is represented by a common generation-investment problem, abstracting from differences among investor-owned utilities, publicly owned utilities, and merchant generators, as well as between vertically integrated and restructured electricity markets. These institutions can affect financing, cost recovery, exposure to wholesale electricity prices, and the response of investment to federal tax credits. The model captures geographic heterogeneity in renewable resources, demand, the existing generation fleet, and electricity trade, but not heterogeneity in ownership and regulation within regions. Incorporating these institutional differences could therefore change the magnitude and geographic distribution of the investment response.

State clean-electricity policies. The model abstracts from state renewable portfolio standards and clean-electricity standards. State policies can partly substitute for federal incentives when the federal subsidy lapses, reducing the response to federal subsidy availability in states with binding clean-energy requirements. Because these policies vary substantially across states, incorporating them could change both the aggregate response and its spatial distribution.

8 Conclusion

Federal renewable subsidies encourage investment in wind and solar, but their future availability has historically been unstable. This paper quantifies the aggregate consequences of this renewable policy instability in the U.S. economy and electricity sector. At the same expected fiscal cost, a stable subsidy produces 13% more renewable capacity than an unstable subsidy and reduces investment in gas. The resulting reduction in fossil generation avoids $360 billion in climate damages and $46 billion in local air-pollution mortality damages. Accounting for the additional resources devoted to renewable investment, the stable subsidy generates $292 billion in net benefits.

The aggregate cost arises because policy changes affect both when new capacity is built and how existing power plants are used. During subsidy lapses, renewable developers delay investment while electricity demand continues to grow. Existing coal and gas plants generate more in the meantime, accounting for 80% of the cumulative emissions difference, while the remaining 20% comes from additional gas capacity. Renewable investment can resume when the subsidy returns, but gas capacity built during a lapse is irreversible and remains in the generation stock until it depreciates. Changes in policy availability can therefore affect emissions both while the subsidy is unavailable and after it returns.

The effects also differ substantially across regions. The stable subsidy matters most where federal subsidies change the renewable investment decision and existing fossil capacity can fill the generation gap when renewable investment is delayed. The Midwest combines both features and accounts for more than half of the avoided local air-pollution mortality damages. Electricity trade creates spatial spillovers: renewable investment in one region can reduce fossil generation and environmental damages elsewhere. California, for example, reduces gas generation partly by importing additional renewable electricity from other regions in the Western Interconnection.

Perfect-foresight counterfactuals separate the effects of policy switching itself from uncertainty about when policy changes occur. When future subsidy changes are known in advance, firms concentrate renewable investment more strongly around known eligibility windows, and convex adjustment costs make this additional bunching costly. Moving from the switching subsidy under perfect foresight to the stable subsidy is worth $355 billion, while resolving uncertainty about the timing is worth −$63 billion. Most of the loss therefore comes from policy switching itself, and uncertainty about the exact dates of policy changes partially mitigates it.

More broadly, the results show that policy stability matters during the green energy transition, even holding expected government spending fixed. The transition requires large, irreversible investments, while the policies that subsidize those investments can change over time. When investment is irreversible and costly to shift across time, policy changes affect not only how much firms invest, but when they invest, which technologies are built, and how the existing capital stock is used in the meantime. These effects can persist after the subsidy returns and, through trade, spill across regions. The design of transition policy therefore involves not only how large a subsidy governments provide, but also how reliably it is provided over time.

References

Acemoglu, Daron, Philippe Aghion, Leonardo Bursztyn, and David Hemous, "The Environment and Directed Technical Change," *The American Economic Review*, 2012, *102* (1), 131–166.

_ , _ , Lint Barrage, and David Hémous, "Climate Change, Directed Innovation, and Energy Transition: The Long-run Consequences of the Shale Gas Revolution," Working Paper 31657, National Bureau of Economic Research 2023.

_ , Ufuk Akcigit, Douglas Hanley, and William Kerr, "Transition to Clean Technology," *Journal of Political Economy*, 2016, *124* (1), 52–104.

Aldy, Joseph E., Jun Gao, and Kenneth T. Gillingham, "Tax Policy and Power Sector Decarbonization," Working Paper, Harvard Kennedy School 2026. June 12, 2026.

_ , Todd D. Gerarden, and Richard L. Sweeney, "Investment versus Output Subsidies: Implications of Alternative Incentives for Wind Energy," *Journal of the Association of Environmental and Resource Economists*, 2023, *10* (4), 981–1018.

Arkolakis, Costas and Conor Walsh, "Clean Growth," NBER Working Paper 31615, National Bureau of Economic Research 2023.

Barradale, Merrill Jones, "Impact of Public Policy Uncertainty on Renewable Energy Investment: Wind Power and the Production Tax Credit," *Energy Policy*, 2010, *38* (12), 7698–7709.

Bernanke, Ben S., "Irreversibility, Uncertainty, and Cyclical Investment," *The Quarterly Journal of Economics*, 1983, *98* (1), 85–106.

Casey, Gregory, Kyle C. Meng, and Ivan Rudik, "Are Carbon Tariffs Climate Policy?," Working Paper 35307, National Bureau of Economic Research 2026.

_ , Woongchan Jeon, and Christian P. Traeger, "The Macroeconomics of Clean Energy Subsidies," *Journal of Environmental Economics and Management*, 2026, *138*, 103311.

Chen, Luming, "The Dynamic Efficiency of Policy Uncertainty: Evidence from the Wind Industry," Technical Report, University of Michigan 2025.

Dennin, Luke R., Nicholas Z. Muller, Medinat Akindele, and Peter J. Adams, "AP4 and AP4T Integrated Assessment Model: Technical Appendix," Technical Report, Carnegie Mellon University and National Bureau of Economic Research May 2024. Fourth installment of the Air Pollution Emission Experiments and Policy (APEEP) model.

Dixit, Avinash K. and Robert S. Pindyck, *Investment under Uncertainty*, Princeton, NJ: Princeton University Press, 1994.

Energy Policy Institute at the University of Chicago and AP-NORC, "2025 Poll: Americans' Views on Climate Change and Policy in 15 Charts," EPIC/AP-NORC Poll, Energy Policy Institute at the University of Chicago (EPIC) 2025. https://epic.uchicago.edu/news/2025-poll-americans-views-on-climate-change-and-policy-in-15-charts/.

Falk, Nathan and Cameron A. Shelton, "Fleeing a Lame Duck: Policy Uncertainty and Manufacturing Investment in US States," *American Economic Journal: Economic Policy*, 2018, *10* (4), 135–152.

French, Anna and Nicholas Vreugdenhil, "Permits as Real Options: Anticipation and Regulatory Leakage," NBER Working Paper 35794, National Bureau of Economic Research 2026.

Fried, Stephie, "Climate Policy and Innovation: A Quantitative Macroeconomic Analysis," *American Economic Journal: Macroeconomics*, 2018, *10* (1), 90–118.

_ , Kevin Novan, and William B. Peterman, "Climate Policy Transition Risk and the Macroeconomy," *European Economic Review*, 2022, *147*, 104046.

Golosov, Mikhail, John Hassler, Per Krusell, and Aleh Tsyvinski, "Optimal Taxes on Fossil Fuel in General Equilibrium," *Econometrica*, 2014, *82* (1), 41–88.

Gowrisankaran, Gautam, Ashley Langer, and Wendan Zhang, "Policy Uncertainty in the Market for Coal Electricity: The Case of Air Toxics Standards," *Journal of Political Economy*, 2025, *133* (6), 1757–1795.

IEA, "World Energy Investment 2026," Technical Report, International Energy Agency, Paris 2026.

Langer, Ashley and Derek Lemoine, "Designing Dynamic Subsidies to Spur Adoption of New Technologies," *Journal of the Association of Environmental and Resource Economists*, 2022, *9* (6), 1197–1234.

Lemoine, Derek, "Green Expectations: Current Effects of Anticipated Carbon Pricing," *Review of Economics and Statistics*, 2017, *99* (3), 499–513.

Poilly, Céline and Fabien Tripier, "Regional Trade Policy Uncertainty," *Journal of International Economics*, 2025, *155*, 104078.

Ricks, Michael David and Owen Kay, "Time-Limited Subsidies: Optimal Taxation with Implications for Renewable Energy Subsidies," Working Paper 2530, Federal Reserve Bank of Dallas 2025. Forthcoming, Journal of Political Economy.

U.S. Bureau of Economic Analysis, "Gross Domestic Product: Implicit Price Deflator, NIPA Table 1.1.9," FRED series GDPDEF, Federal Reserve Bank of St. Louis, https://fred.stlouisfed.org/series/GDPDEF 2024. Index 2017 = 100; annual averages 2020 = 105.4, 2022 = 118.0.

U.S. Energy Information Administration, "Form EIA-923, Power Plant Operations Report," https://www.eia.gov/electricity/data/eia923/ 2017. Annual net generation and fuel consumption by plant.

U.S. Environmental Protection Agency, "Report on the Social Cost of Greenhouse Gases: Estimates Incorporating Recent Scientific Advances," Technical Report, U.S. Environmental Protection Agency 2023.

A Additional model details

A.1 Subsidy eligibility

The wind and solar eligibility indicators are

$$\chi_t^W = \mathbf{1}[s_t = 1], \qquad \chi_t^S = \mathbf{1}[t < 2028] + \mathbf{1}[t \geq 2028]\mathbf{1}[s_t = 1]. \tag{A.1}$$

Wind invested while the subsidy is active enters the eligible vintage $K_{jt}^{W,PTC}$ according to (4.4). The production tax credit is grandfathered. Solar receives the investment tax credit at construction.

A.2 Electricity prices and profits

Substituting the Pareto survival function (4.10) into the expected-price expression (4.12) gives the wholesale price in closed form. Define

$$x_{jt}^{d2} = Q_{jt}^0, \qquad x_{jt}^{d3} = Q_{jt}^0 + K_{jt}^{\text{lo}}, \qquad \bar{Q}_{jt} = Q_{jt}^0 + K_{jt}^{\text{lo}} + K_{jt}^{\text{hi}}.$$

Then

$$\begin{aligned} \bar{p}_{jt}^E = mc_{jt}^{\text{lo}} &\left(\frac{x_{jt}^{\min}}{x_{jt}^{d2}}\right)^{\theta_j} \\ &+ (mc_{jt}^{\text{hi}} - mc_{jt}^{\text{lo}}) \left(\frac{x_{jt}^{\min}}{x_{jt}^{d3}}\right)^{\theta_j} \\ &+ (V_j - mc_{jt}^{\text{hi}}) \left(\frac{x_{jt}^{\min}}{\bar{Q}_{jt}}\right)^{\theta_j}, \end{aligned} \tag{A.2}$$

with each term capped at its coefficient when the corresponding threshold lies below $x_{jt}^{\min}$. Additional zero-marginal-cost capacity raises the dispatch thresholds and lowers the expected wholesale price.

Regional net exports are

$$X_{jt} = G_{jt} - \bar{E}_{jt}.$$

The local price is related to the interconnection hub price by (4.13). The associated trans-

mission cost,

$$c_j(X_{jt}) = \frac{\gamma}{2}\frac{X_{jt}^2}{\overline{TL}_j},$$

is a real resource cost. The congestion basis itself is a transfer and is returned to households through $\mathcal{R}_{jt}$.

Electricity-sector profit is

$$\begin{aligned}\Pi_{jt}^E = p_{jt}^E \left[(CF_j^W K_{jt}^W + CF_j^S K_{jt}^S + Q_{jt}^{\text{base}})8760 + gen_{jt}^G + gen_{jt}^C\right] \\ + \phi^{PTC} CF_j^W 8760 K_{jt}^{W,PTC} - mc_t^G gen_{jt}^G - mc_t^C gen_{jt}^C.\end{aligned} \tag{A.3}$$

The first line is electricity-market revenue; the second adds the production tax credit earned by eligible wind and subtracts fossil fuel costs.

A.3 Capacity adequacy

I impose a capacity-adequacy requirement to ensure that each region maintains enough reliable generation capacity to meet forecast demand. Not all installed capacity contributes equally to system reliability. Gas, coal, nuclear, and hydro count fully toward the requirement, while wind and solar count according to their effective load-carrying capability. Accredited capacity is therefore

$$K_{jt}^{acc} = K_{jt}^G + K_{jt}^C + Q_{jt}^{base} + \zeta_j^W K_{jt}^W + \zeta_j^S K_{jt}^S, \tag{A.4}$$

where ζ_j^W and ζ_j^S measure the contribution of wind and solar capacity to system reliability. For example, $\zeta_j^W = 0.15$ means that one MW of installed wind capacity contributes 0.15 MW toward the capacity-adequacy requirement.

The system must satisfy

$$\Pr\left(D_{jt}^{fc} > K_{jt}^{acc}\right) \leq \varepsilon, \tag{A.5}$$

where D_{jt}^{fc} is exogenous forecast load and ε is the target probability that forecast load exceeds accredited capacity. Because the requirement is based on exogenous forecast demand, it does not respond to the model's endogenous electricity price or electricity demand.

Under the Pareto load distribution, (A.5) implies a minimum level of accredited capacity in each region. Because gas is the only dispatchable generation technology that can be built endogenously, gas investment fills any shortfall when existing accredited capacity falls below this requirement.

The multiplier on the capacity-adequacy constraint is the capacity price p_{jt}^{cap}. When the constraint binds,

$$p_{jt}^{cap} = \max\left\{p_t^G(r_t^K + \delta^G) - \mathrm{MPK}_{jt}^{G,\mathrm{energy}},\, 0\right\}. \tag{A.6}$$

The capacity price is the shadow value of additional reliable generation capacity. It equals the difference between the user cost of gas capacity and the return that gas earns in the energy market—the "missing money" required to support enough gas investment to satisfy the capacity-adequacy constraint. Because p_{jt}^{cap} is a shadow value rather than a resource cost, it does not enter aggregate welfare. Each region's capacity-adequacy requirement is determined by its own forecast demand.

A.4 Formal equilibrium definition

Within-period equilibrium. Given generation-capital stocks $\mathbf{K}_{jt}$, policy state s_t, and the exogenous demand and cost paths, a within-period equilibrium consists of regional electricity prices $\{p_{jt}^E\}$, interconnection hub prices $\{\bar{p}_{mt}^E\}$, generation and dispatch, household and firm electricity demands, and net trade such that:

1. each region dispatches generation in merit order and the wholesale price satisfies (4.12);
2. household and firm electricity demands satisfy (4.11);
3. local and hub prices satisfy (4.13);
4. electricity markets clear within each interconnection; and
5. congestion-basis payments are returned through $\mathcal{R}_{jt}$.

Dynamic equilibrium. A dynamic equilibrium is a within-period equilibrium in every period together with capital stocks, investment policies, financial wealth, consumption, and the national capital rental such that:

1. generation investment solves (4.5) subject to irreversibility, capital accumulation (4.3), and the wind subsidy vintage (4.4);
2. capital evolves according to its laws of motion;
3. the stochastic discount factor is consistent with household consumption;
4. the national production-capital market satisfies (4.9);
5. regional labor markets and the aggregate goods market clear;
6. the federal government budget balances;
7. the policy state follows Π; and
8. the economy converges to the terminal stochastic steady state.

The federal government finances subsidy outlays through the labor-income tax and national lump-sum transfers, and separately returns interregional congestion revenue to households. Its period budget balances:

$$\sum_j \varsigma_t w_{jt} L_{jt} + \Xi_t^{\text{cong}} = \Phi_t + \sum_j \mathcal{R}_{jt} + \sum_j N_j T_t^{nat}, \tag{A.7}$$

where $\Phi_t = \sum_j (\text{PTC}_{jt} + \text{ITC}_{jt})$ is aggregate subsidy expenditure, Ξ_t^{cong} is congestion revenue, and the rebate returns it, $\Xi_t^{\text{cong}} = \sum_j \mathcal{R}_{jt}$.

A.5 Resource accounting

By aggregate goods-market clearing, the consumption component of welfare in (4.15) can equivalently be written

$$W^C = \sum_t \beta^{t-2010} \Bigg[\sum_j Y_{jt} + \sum_j p_{jt}^E E_{jt}^{HH} - \sum_{j,k} p_t^k I_{jt}^k - \sum_{j,k} \Psi^k(I_{jt}^k, K_{jt}^k) \\ - \sum_j \left(mc_t^G gen_{jt}^G + mc_t^C gen_{jt}^C \right) - \sum_j c_j(X_{jt}) - \Delta K_{P,t}^{\text{total}} \Bigg]. \tag{A.8}$$

Generation investment, adjustment costs, fossil fuel use, transmission, and production-capital investment are real resource costs. Household electricity is valued at its market

price. Generation investment is valued at its full resource price p_t^k, rather than the ITC-reduced private price $\tilde{p}_t^k$, so that the ITC remains a transfer rather than appearing as a reduction in resource cost. The production-capital term is *gross* investment, $\Delta K_{P,t}^{\text{total}} \equiv \sum_j \left(A_{j,t+1} - (1-\delta_P)A_{jt}\right) = \sum_j S_{jt}$, so that depreciation replacement is counted exactly once.

Why subsidies and their financing cancel. That federal subsidies, the taxes that finance them, and congestion rebates are transfers is a consequence of the balanced federal budget (A.7), not a separate assumption. The wind credit PTC_{jt} is paid inside Π_{jt}^E and the solar credit $\text{ITC}_{jt} = \phi^{ITC}\chi_t^S p_t^S I_{jt}^S$ is embedded in the after-subsidy price, so $\tilde{p}_t^S = p_t^S - \text{ITC}$ per unit and $\Pi_{jt}^E = \Pi_{jt}^{E,\text{ex-PTC}} + \text{PTC}_{jt}$. Substituting the government budget (A.7) into the household budget (4.2), the subsidy expenditure Φ_t, the labor tax ς_t, and the congestion rebate $\mathcal{R}_{jt}$ cancel, leaving the household's effective budget in full resource prices:

$$C_{jt} + p_{jt}^E E_{jt}^{HH} + \sum_k p_t^k I_{jt}^k + \sum_k \Psi^k + S_{jt} = w_{jt}L_{jt} + r_t^K A_{jt} + \Pi_{jt}^{E,\text{ex-PTC}}. \tag{A.9}$$

The labor tax cancels from the resource budget (A.9) for any labor-supply elasticity; it is *immaterial* for allocations because labor is inelastic, so the wedge $(1-\varsigma_t)$ distorts no margin. With elastic labor the tax would still cancel from the budget but would lower labor supply through the after-tax wage, and ς_t would then be material. The subsidy survives only in the investment first-order conditions (4.6)–(4.7), through the subsidized prices $\tilde{p}_t^k$ and the PTC stream. Summing (A.9) across regions and using goods-market clearing delivers the resource form (A.8).

B Derivation of the recursive competitive equilibrium

This appendix derives the household's optimality conditions, establishes the Fisher separation that justifies stating generation investment as the value-maximization (4.5), records the envelope conditions behind the investment first-order conditions, and defines the stationary recursive competitive equilibrium that serves as the model's terminal condition. It supplies the derivations behind the equilibrium definition in Appendix A.4.

B.1 Household optimality and Fisher separation

The household in region j chooses consumption C_{jt}, electricity E^{HH}_{jt}, saving S_{jt}, and generation investment $\{I^k_{jt}\}_{k\in\{S,W,G\}}$ to maximize (4.1) subject to the budget constraint (4.2), the financial-wealth law of motion $A_{j,t+1} = (1-\delta_P)A_{jt} + S_{jt}$, and the generation-capital accumulation (4.3)–(4.4). It holds two stores of value: financial wealth A_{jt}, which earns the national rental r^K_t, and generation capital $\mathbf{K}_{jt}$, which it owns and operates.

Static conditions. The intratemporal first-order conditions for C_{jt} and E^{HH}_{jt} deliver the marginal utility of the numeraire, $\lambda_{jt} = \beta^t \mathcal{U}_C(C_{jt}, E^{HH}_{jt})$, and the electricity-demand schedule (4.11), $E^{HH}_{jt} = \bar{\alpha}_h C_{jt}(p^E_{jt})^{-\sigma}$.

The saving Euler equation. The first-order conditions for S_{jt} and $A_{j,t+1}$ combine to

$$1 = \mathbb{E}_t\big[\mathcal{M}_{t,t+1}\,(1-\delta_P + r^K_{t+1})\big], \qquad \mathcal{M}_{t,t+1} = \beta\,\frac{\mathcal{U}_C(C_{j,t+1}, E^{HH}_{j,t+1})}{\mathcal{U}_C(C_{jt}, E^{HH}_{jt})}, \tag{B.1}$$

which both defines the stochastic discount factor and prices the return on financial wealth.

The marginal utility of consumption follows from the CES felicity (4.1). With period felicity $\log\mathcal{U}(C,E)$ and $\partial\mathcal{U}/\partial C = (\mathcal{U}/C)^{1/\sigma}$,

$$\frac{\partial \log \mathcal{U}}{\partial C} = \frac{1}{\mathcal{U}}\frac{\partial \mathcal{U}}{\partial C} = C^{-1/\sigma}\,\mathcal{U}^{(1-\sigma)/\sigma},$$

so the stochastic discount factor is, exactly,

$$\mathcal{M}_{t,t+1} = \beta\left(\frac{C_{jt}}{C_{j,t+1}}\right)^{1/\sigma}\left(\frac{\mathcal{U}_{j,t+1}}{\mathcal{U}_{jt}}\right)^{(1-\sigma)/\sigma}. \tag{B.2}$$

As electricity's expenditure share goes to zero, $\mathcal{U} \to C$ and the two exponents sum to one ($\frac{1}{\sigma} - \frac{1-\sigma}{\sigma} = 1$), so $\mathcal{M}_{t,t+1} \to \beta\, C_{jt}/C_{j,t+1}$. With $\sigma = 0.3$ the individual exponents are large, but electricity is roughly 2% of expenditure and co-moves with consumption, so the two factors nearly cancel; the solver uses the limiting log-in-consumption form $\mathcal{M}_{t,t+1} = \beta\, C_{jt}/C_{j,t+1}$.

Fisher separation. The first-order condition for generation investment $I^k_{jt} > 0$ equates its marginal cost to the discounted marginal value of installed capital,

$$\tilde{p}^k_t + \Psi^k_I(I^k_{jt}, K^k_{jt}) + \nu^k_{jt} = \mathbb{E}_t\big[\mathcal{M}_{t,t+1}\, q^k_{j,t+1}\big], \qquad q^k_{jt} \equiv \frac{\partial V_t}{\partial K^k_{jt}}, \tag{B.3}$$

where $\nu^k_{jt} \geq 0$ is the multiplier on irreversibility $I^k_{jt} \geq 0$, with complementary slackness $\nu^k_{jt} I^k_{jt} = 0$, and for wind and solar $\tilde{p}^k_t$ is replaced by the regional site-cost schedule $c^k_j(\cdot)$. The budget multiplier λ_{jt} cancels from both sides, so (B.3) depends on the household's consumption path only through the common discount factor $\mathcal{M}_{t,t+1}$. Generation investment therefore maximizes the present value of generation profits at the market discount factor, independent of the consumption and saving choice. This is Fisher separation: with a frictionless national capital market the investment problem can be stated as the value-maximization (4.5), while household consumption is pinned separately by (4.11), (B.1), and the budget constraint. Consequently V_t in (4.5) is the value of the generation business, not household welfare; welfare is the utility sum (4.1), evaluated in resource terms in Appendix A.5.

Regional consumption allocation. The national capital market is frictionless and the rental r^K_t and discount factor β are common across regions, so the saving Euler (B.1) implies a *common* consumption growth rate, $C_{j,t+1}/C_{jt} = \beta(1 - \delta_P + r^K_{t+1})$, with no region index on the right-hand side. Regional consumption paths are therefore parallel, and the level is pinned by region j's lifetime budget at the market discount factor:

$$\frac{C_{j,2010}}{1-\beta} = A_{j,2010} + \sum_t Q_t\, y_{jt}, \qquad Q_t = \prod_{s \leq t} \big(1 - \delta_P + r^K_s\big)^{-1}, \tag{B.4}$$

where y_{jt} is the region's net resource flow from the effective budget (A.9). Consumption equals the permanent income of the regional resource flow discounted at the market rate, and the regional financial-wealth path A_{jt} is a determined residual that does not enter welfare beyond C_{jt}. This is the closed-form solution of the regional saving problem under the national capital market; it coincides with explicitly tracking A_{jt} and clearing a national bond market period by period, and reduces to discounting at β only when r^K_t equals the golden-rule rate $1/\beta - 1 + \delta_P$ in every period.

B.2 Investment first-order and envelope conditions

Applying the envelope theorem to (4.5), the marginal value of each capital stock obeys

$$q^W_{jt} = \Pi^E_{K^W,jt} + (1-\delta^W)\,\mathbb{E}_t[\mathcal{M}_{t,t+1} q^W_{j,t+1}], \tag{B.5}$$

$$q^{W,PTC}_{jt} = \Pi^E_{K^{W,PTC},jt} + (1-\delta^W)(1-h^{exit})\,\mathbb{E}_t[\mathcal{M}_{t,t+1} q^{W,PTC}_{j,t+1}], \tag{B.6}$$

$$q^S_{jt} = \Pi^E_{K^S,jt} + (1-\delta^S)\,\mathbb{E}_t[\mathcal{M}_{t,t+1} q^S_{j,t+1}], \tag{B.7}$$

$$q^G_{jt} = \Pi^E_{K^G,jt} + p^{cap}_{jt} + (1-\delta^G)\,\mathbb{E}_t[\mathcal{M}_{t,t+1} q^G_{j,t+1}], \tag{B.8}$$

where $\Pi^E_{K^k,jt} = \partial \Pi^E_{jt} / \partial K^k_{jt}$ is the marginal electricity profit of capital k from (A.3) and p^{cap}_{jt} is the capacity-adequacy shadow value (A.6), nonzero only when the reserve constraint (A.5) binds. Combining (B.3) with (B.5)–(B.6) and the vintage rule $\partial K^{W,PTC}_{j,t+1} / \partial I^W_{jt} = \mathbf{1}[s_t{=}1]$ yields the wind condition (4.6); combining it with (B.7) and the solar site-cost curve (5.1) yields the solar condition (4.7), and the gas condition follows from (B.8). The marginal energy value $\Pi^E_{K^W,jt}$ falls with K^W_{jt} because renewable entry lowers the price wind captures (cannibalization), while the marginal cost $c^W_j(K^W_{jt})$ rises with K^W_{jt} as a region exhausts its best sites, so the wind condition admits a unique interior stock.

B.3 The stationary recursive competitive equilibrium

The model's terminal condition is a stationary recursive competitive equilibrium. Fundamentals (capital-cost paths, electricity demand, coal retirement) are held at their terminal-year values, so the environment is time-invariant and decision rules depend on the state $(\mathbf{K}, s)$ alone.

Definition. A stationary recursive competitive equilibrium is a value function $V(\mathbf{K}, s)$, an investment policy $I(\mathbf{K}, s)$, regional and hub prices $\{p^E_j(s), \bar{p}^E_m(s)\}$, a national capital rental r^K, and a distribution $\mu(\mathbf{K}, s)$ over the state such that:

1. V and I solve the stationary Bellman

$$V(\mathbf{K}, s) = \max_{I \geq 0} \Big\{ \Pi^E(\mathbf{K}, s) - \Xi^S(I^S) - \Xi^W(I^W) - \tilde{p}^G I^G - \textstyle\sum_k \Psi^k + \mathbf{1}[s{=}1]\, \bar{\mathcal{M}}\, q^{WPTC} I^W + \sum_{s'} \Pi_{ss'} \mathcal{M}_{ss'} V(\mathbf{I}$$

2. prices clear the within-period electricity and interconnection-trade equilibrium of Appendix A.4 in each state s;

3. the saving Euler (B.1) holds in stationary form, $1 = \sum_{s'} \Pi_{ss'} \mathcal{M}_{ss'}(1 - \delta_P + r^K)$; and

4. μ is invariant under the policy and the policy Markov,

$$\mu(\mathbf{K}', s') = \sum_{s} \Pi_{ss'} \int_{\{\mathbf{K} : (1-\delta)\mathbf{K} + I(\mathbf{K}, s) = \mathbf{K}'\}} \mu(d\mathbf{K}, s). \tag{B.9}$$

Permanent subsidy: a deterministic point. When the forward process is absorbing in the active state, $\Pi = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right)$, the problem collapses to a single state and the invariant distribution (B.9) is a point mass at $\mathbf{K}^\star$ with zero net investment, $I^k(\mathbf{K}^\star) = \delta^k K_k^\star$. Substituting the envelope condition (B.5) into the investment condition (B.3) at $I^W = \delta^W K^W$ gives the scalar fixed point for the stationary wind stock,

$$\underbrace{c_j^W(K^{W\star}) + \psi^W \delta^W}_{\text{marginal cost, increasing in } K^W} = \underbrace{\frac{\beta\, \Pi_{K^W}^E(K^{W\star})}{1 - \beta(1 - \delta^W)}}_{\text{energy value, decreasing in } K^W} + \underbrace{\beta\, q^{WPTC}}_{\text{subsidy wedge}}. \tag{B.10}$$

The left side increases and the right side decreases in K^W, so (B.10) has a unique interior solution; the permanent subsidy shifts the right side up by βq^{WPTC} at every K^W, raising $K^{W\star}$.

Unstable subsidy: an ergodic distribution. Under the estimated Markov Π both states are recurrent, and the invariant distribution $\mu^\star(\mathbf{K}, s)$ is non-degenerate: capital cycles with the policy state and the long-run stock is the ergodic mean $\mathbb{E}_{\mu^\star}[\mathbf{K}]$. The subsidy term in (B.3) enters only in $s = 1$, and irreversibility binds in lapses, $\nu^W(0) > 0$ with $I^W(0) = 0$, so no subsidy-dependent wind is built while the subsidy is off. The ergodic mean therefore satisfies

$$\mathbb{E}_{\mu^\star}[K^W] < K^{W\star}, \tag{B.11}$$

strictly below the permanent steady state. The two steady states are the same equilibrium operator evaluated at different Π: the permanent point $\mathbf{K}^\star$ is the absorbing-state limit of the stochastic ergodic distribution, and any Π with positive long-run mass on the lapsed state yields a strictly lower renewable stock and, through the capacity-adequacy margin (A.4), a higher gas stock.

C Computational algorithm

The model is a non-stationary stochastic spatial general equilibrium. I solve it by nesting the regional dynamic investment problems inside the within-period electricity-market and general-equilibrium fixed points.

C.1 Terminal stochastic steady state

Backward induction requires a terminal condition. I solve the stationary value function at terminal fundamentals by value-function iteration on a three-dimensional capital grid (K^S, K^W, K^G) crossed with the two policy states. At each grid point, the three investment choices $(I^S, I^W, I^G) \geq 0$ are optimized by coordinate-ascent golden-section search.

The eligible wind vintage $K^{W,PTC}$ is not an additional grid dimension. Conditional on eligibility, the production credit is separable from the energy value of wind, so its contribution can be represented by the present value

$$q^{W,PTC} = \frac{\phi^{PTC} CF_j^W \,\mathrm{HRS}}{1 - \beta(1-\delta^W)(1-h^{exit})}. \tag{C.1}$$

This value enters the marginal return to wind investment only when new capacity qualifies for the subsidy.

C.2 Backward-induction transition

Given the terminal value function, I solve the finite-horizon transition from 2010 to 2130 backward with time-varying fundamentals; quantitative results are reported through 2070.

C.3 Within-period and general-equilibrium fixed points

Within each period, a low-dimensional fixed point jointly determines price-elastic electricity demand, merit-order prices, and the scale of the Pareto load distribution. Interconnection hub prices clear net trade among member regions. An outer fixed point jointly determines state-contingent local electricity prices, the national capital rental, and aggregate consump-

tion.

D Illustration of the investment mechanism

I solve the full model for the Eastern Interconnection along a deterministic policy path in which the subsidy is active, lapses for six years, and is then permanently restored. Figure 11 follows the Midwest, where the federal subsidy is important for marginal wind investment.

While the subsidy is active, Midwest wind investment runs at approximately 11 GW per year. When the subsidy lapses, newly built wind no longer qualifies for the PTC and wind investment falls to zero.

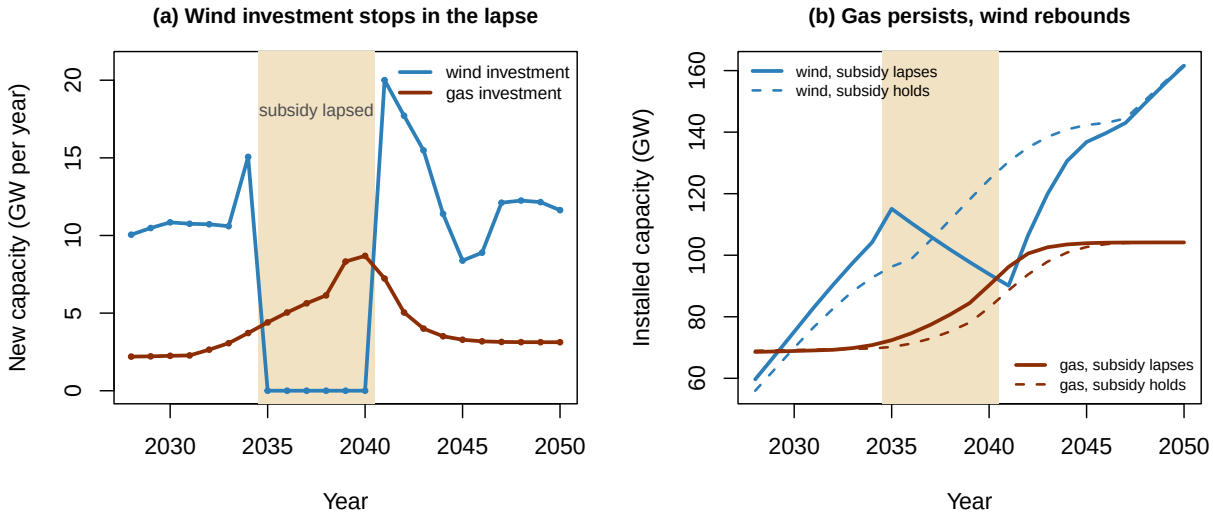


Figure 11: Investment and capacity around a temporary subsidy lapse.
Notes. Full model solved for the Eastern Interconnection along a deterministic policy path in which the subsidy is active, lapses over 2035–2040, and is then restored (the Midwest).

E Policy-process calibration

This appendix reports the pre-2028 policy beliefs and the political transition matrix used to construct the policy process in Section 5.3.

E.1 Pre-2028 policy path

Table 9: Common pre-2028 policy path.

Years	Belief	Pr(lapse | active)	Event
2010	active	—	initial state $s_0 = 1$
2010–2021, off-cliff	active, low lapse risk	0.08	
2012, 2014, 2016, 2019, 2021	scheduled expiration	0.60	PTC expiration dates
2022–2027	active	0.00	IRA commitment
2027	lapse in 2028	1.00	OBBBA repeal
2028 onward	regime-specific	—	counterfactual branch

Notes. The pre-2028 path is common to every counterfactual. The counterfactual policy processes begin in 2028.

E.2 Federal political control

I group each Congress from 1993 through 2025 as a Democratic unified control, Republican unified control, or divided government. Table 10 reports the transition frequencies.

Table 10: Federal political-control transitions, 1993–2025.

	Next Congress		
Current Congress	Democratic trifecta	Republican trifecta	Divided
Democratic trifecta	0.00	0.00	1.00
Republican trifecta	0.00	0.33	0.67
Divided	0.20	0.30	0.50

Notes. Transition frequencies are calculated across two-year Congresses. The ergodic frequencies of Democratic trifecta, Republican trifecta, and divided government are 0.12, 0.27, and 0.61, respectively.

F Calibration and data

Table 11 reports regional summary statistics, and Figure 12 plots the principal exogenous paths.

Table 11: Summary statistics: the U.S. wind, solar, and gas sector by region.

Region	Inter-connect	Wind CF	Solar CF	Wind (GW)	Solar (GW)	Gas (GW)	Demand (TWh)
Southwest	West	0.38	0.32	5.7	7.6	20.7	117
Central	East	0.37	0.26	28.8	0.6	24.8	263
Midwest	East	0.34	0.21	41.3	15.5	98.0	667
Texas	ERCOT	0.34	0.29	42.3	22.5	64.9	403
Northwest	West	0.34	0.29	20.0	12.0	29.9	347
Carolinas	East	0.28	0.25	0.4	8.4	25.6	214
California	West	0.27	0.31	6.5	22.3	34.6	249
Mid-Atlantic	East	0.27	0.19	3.7	11.2	90.3	751
New England	East	0.26	0.19	1.6	3.1	15.8	112
New York	East	0.23	0.19	2.9	2.7	20.7	140
Tennessee	East	0.10	0.25	0.0	0.6	6.9	186
Southeast	East	0.04	0.27	0.0	5.7	35.3	227
Florida	East	0.04	0.27	0.0	10.9	54.6	241
U.S. total				153	123	522	3,919

Notes. Capacity factors are net generation over nameplate capacity, 2019–2024 average (EIA-923/EIA-860); regions with negligible wind show 0. Installed capacity is 2024 (EIA-860). Demand is 2024 electricity sales (AEO 2025). Sources: EIA-860, EIA-923, EIA AEO 2025.

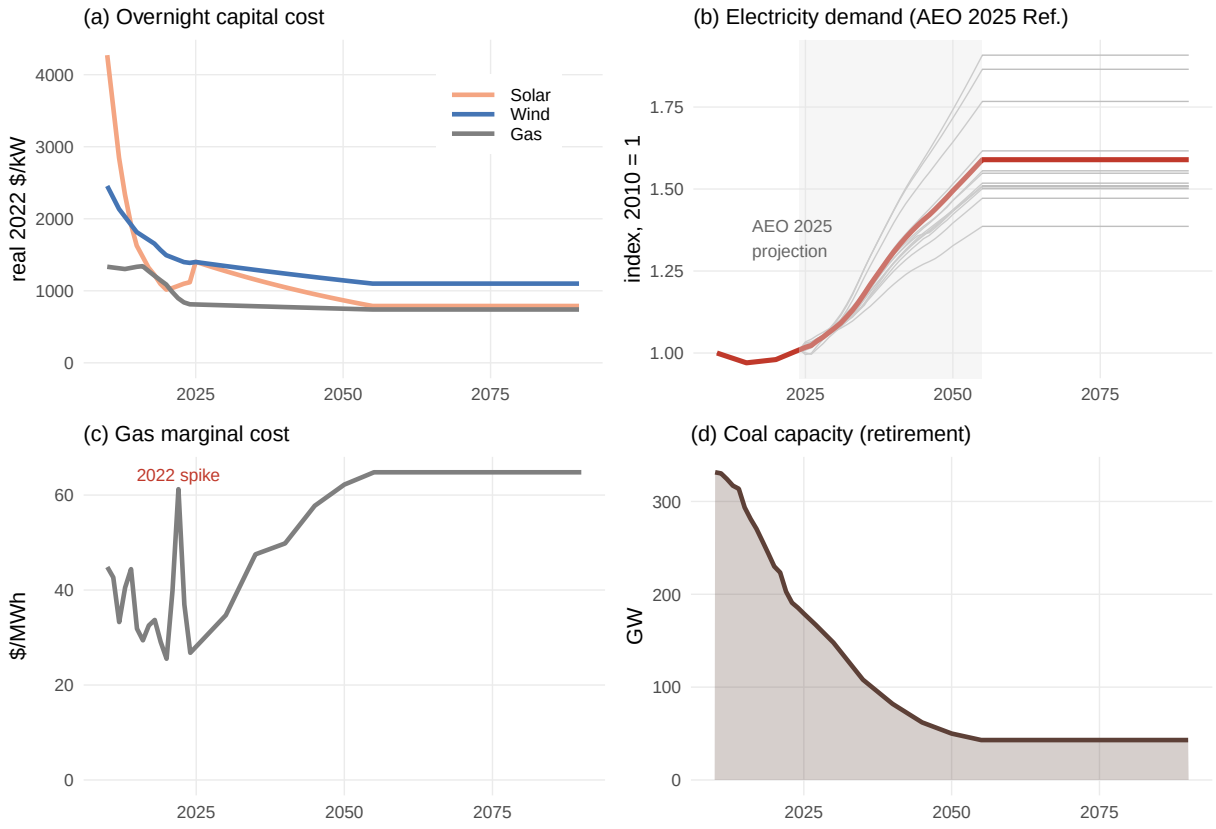


Figure 12: The exogenous processes that drive the transition.

Notes. Panel (a): overnight capital costs (real 2022 $/kW). Panel (b): electricity-demand index (2010=1). Panel (c): gas marginal cost ($/MWh). Panel (d): coal capacity (GW). Sources: NREL ATB, LBNL/EIA, and EIA AEO 2025.

F.1 Capacity, generation, and capacity factors

Initial generation stocks and the historical coal-retirement path come from EIA-860 and are aggregated from generators to the 13 model regions by technology. After the observed period, national coal retirements follow the EIA AEO path and are allocated across regions.

Regional capacity factors are calculated as the ratio of EIA-923 net generation by EIA-860 nameplate capacity (averaged over 2019–2024):

$$CF_j^k = \frac{\text{net generation}}{\text{nameplate capacity} \times 8760}.$$

Gas is represented using the combined-cycle capacity factor. Fossil marginal costs combine delivered fuel prices with heat rates and transition toward the AEO 2025 path after the historical period.

F.2 Demand and hourly load

Regional annual electricity demand follows the EIA AEO 2025 Reference case, cross-walked to the 13 model regions. For each region, I estimate the Pareto tail parameter θ_j by maximum likelihood using the upper tail of hourly demand (EIA-930).

F.3 Prices, trade, and emissions

The regional trade slope is

$$s_j = \frac{\gamma}{\overline{TL}_j}.$$

For regions with observed hub prices, I estimate the slope using the average wholesale-price basis and average net position. For the remaining regions, I impute it from a common congestion parameter γ and observed gross interchange capacity. Net positions and interchange are from EIA-930. Texas is treated as a separate interconnection with no trade with the other model regions.

Emissions factors are $\xi_G = 0.40$ and $\xi_C = 0.95$ tCO_2/MWh from EPA eGRID.

F.4 Federal tax-credit history

Figure 13 summarizes the statutory history used to construct the historical policy path.

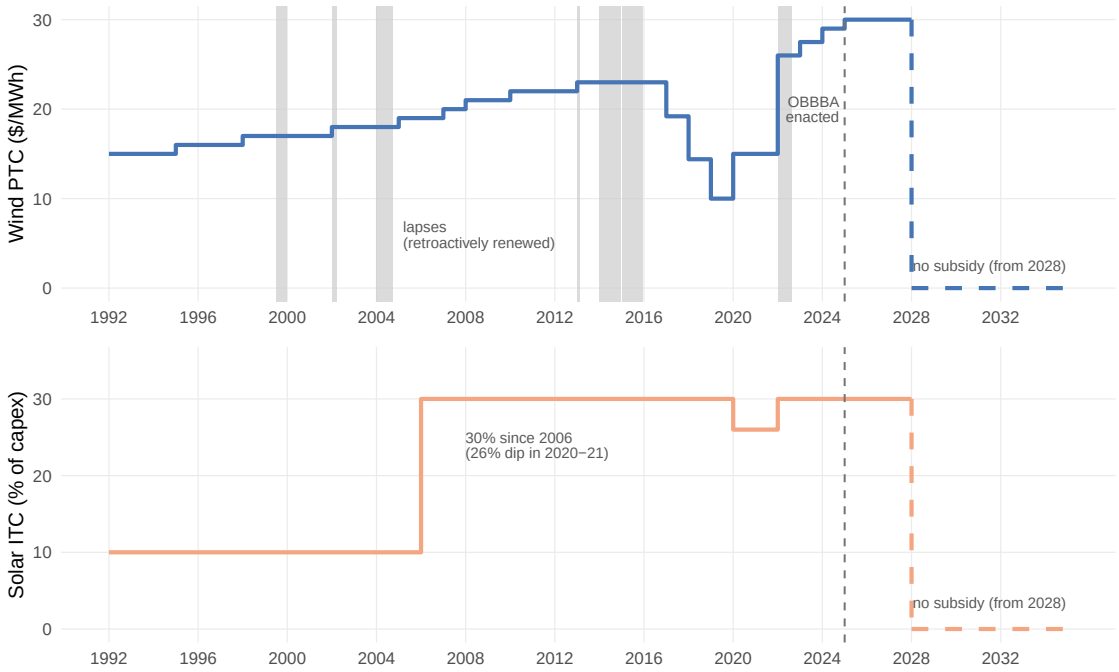


Figure 13: Federal wind and solar tax-credit history and scheduled termination, 1992–2035.

Notes. Source: enacting statutes and IRS notices. The wind PTC and solar ITC remain available for projects placed in service through 2027 under the 2025 One Big Beautiful Bill Act; the dashed segments show the scheduled termination from 2028. Projects beginning construction by mid-2026 may remain eligible if placed in service by 2030 under safe-harbor rules.

Table 12: The production tax credit in the economics of a representative wind project.

	$/MWh
Revenue per MWh, during the ten-year credit period	
Wholesale energy price	50
Production tax credit (§45)	26
Total	76
Levelized cost per MWh, unsubsidized	
Capital ($1,600/kW, 25 yr, 5%)	39
Fixed O&M ($40/kW-yr)	14
Levelized cost of energy	53
Net profit per MWh	
Without the credit	−3
With the credit	+23
Credit as a share of revenue	34%

Notes. All entries are dollars per megawatt-hour generated.

Table 13: Interconnection trade slope by region.

Region	Interconnection	X_j (GW)	$\overline{TL}_j$ (GW)	s_j	Source
Mid-Atlantic	Eastern	+5.9	7.9	0.24	direct
Central	Eastern	+1.1	1.1	1.79	imputed
Southeast	Eastern	+1.0	2.0	1.00	imputed
New England	Eastern	-0.2	0.2	8.56	imputed
Carolinas	Eastern	-0.3	1.1	1.80	imputed
Tennessee	Eastern	-0.4	3.0	0.68	imputed
Florida	Eastern	-0.4	0.4	4.98	imputed
New York	Eastern	-1.6	2.1	0.99	imputed
Midwest	Eastern	-2.7	6.0	3.58	direct
Northwest	Western	+3.5	3.9	0.52	direct
Southwest	Western	+2.1	4.0	0.52	imputed
California	Western	-6.0	6.0	0.34	imputed
Texas	Texas	-0.2	0.2	12.80	autarky

Notes. The trade slope $s_j = \gamma/\overline{TL}_j$ is the wholesale price basis a region pays per GW of net exports X_j (positive = net exporter). For regions with hub prices (ICE, and NYISO for New York), s_j is estimated *directly* as the mean price basis over the mean net position; for the others it is *imputed* as $\gamma/\overline{TL}_j$ from a common congestion constant $\gamma \approx 2$ (the median of $s_j\overline{TL}_j$ across the priced regions) and the gross-interchange proxy $\overline{TL}_j$. Sources: ICE, NYISO, EIA-930.

G Additional model validation

Section 5.4 reports the principal untargeted moments. This appendix provides the corresponding dynamic and regional comparisons.

G.1 Capacity

Figure 14 compares modeled and observed national wind and solar capacity over 2010–2024 and the 2024 regional cross-section. National wind and solar capacity levels enter the internal calibration; their allocation across regions does not. The model reproduces the regional ordering of wind capacity particularly closely, with a Spearman rank correlation of 0.96, while the corresponding solar correlation is 0.57.

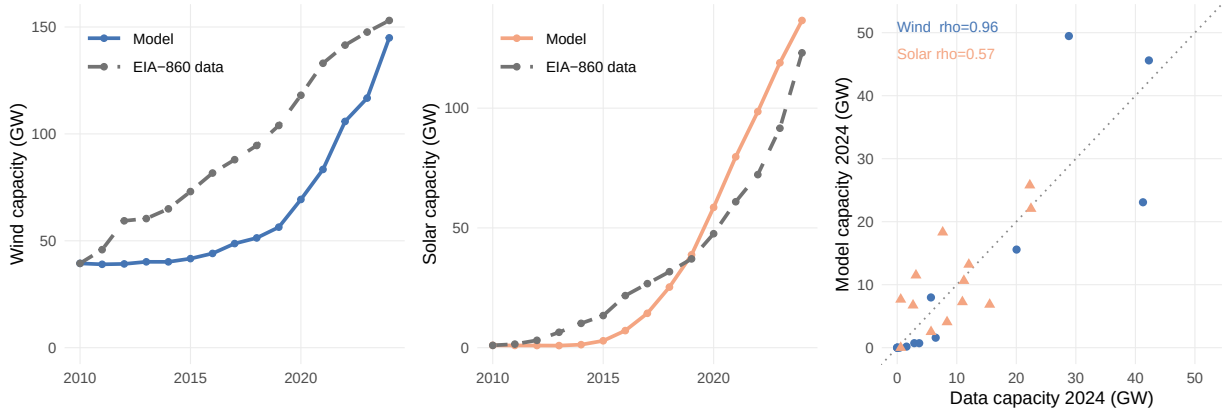


Figure 14: Model versus data: capacity, 2010–2024.

Notes. Panels (a)–(b) compare national wind and solar capacity in the model and EIA-860 data. Panel (c) compares modeled and observed 2024 capacity across the 13 regions. Source: EIA-860.

G.2 Wholesale electricity prices

Figure 15 compares modeled wholesale prices with observed trading-hub prices by interconnection. The model reproduces the broad price level across interconnections and ERCOT as the low-price region.

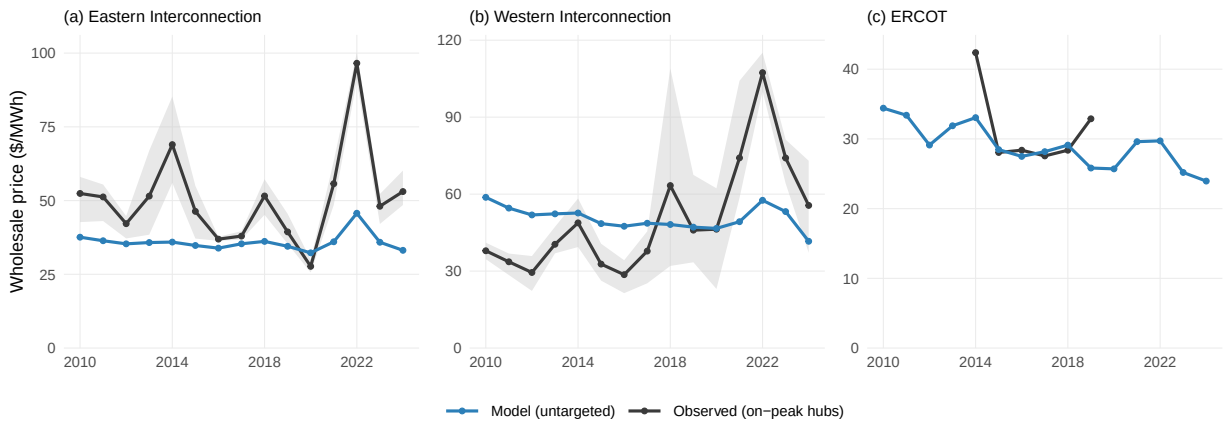


Figure 15: Untargeted wholesale price, by interconnection.

Notes. The black line is the mean across hubs and the grey band is the range. Eastern hubs are PJM West, the Indiana Hub, and the Mass Hub; western hubs are Mid-C, NP15, SP15, and Palo Verde. Source: EIA wholesale market data (ICE hub prices).

G.3 Local air-pollution damages

Source damages. I use AP4, the fourth installment of the Air Pollution Emission Experiments and Policy model (Dennin et al., 2024), with a 2017 emissions and population base. AP4 treats each electric generating unit as a point source with its own effective-stack-height source–receptor matrix and computes the marginal mortality damage, in dollars per metric ton, of its NH_3, NO_x, primary $PM_{2.5}$, SO_2, and VOC emissions through their contribution to ambient fine-particulate concentrations. The endpoint is $PM_{2.5}$-related mortality, valued at the value of a statistical life; "$PM_{2.5}$-related mortality" refers to this health endpoint, mortality from ambient fine-particulate exposure, and not only to direct emissions of $PM_{2.5}$. A source damage is the total contiguous-U.S. mortality damage from one additional metric ton emitted at that source. AP4 monetary outputs are in real 2020 dollars.

Matching to generation. AP4 generating units are identified by ORIS plant code. I match them to EIA-923 net generation (U.S. Energy Information Administration, 2017) by ORIS, aggregating AP4 point sources to the ORIS plant level. I classify plants as coal or natural gas using EIA fuel and technology information.

Plant-level damages. For plant i and pollutant p, let MD_{ip} be the AP4 marginal damage (dollars per metric ton), E_{ip} observed annual emissions (metric tons), and G_i net generation

(MWh). The plant damage rate, in dollars per MWh, is

$$d_i = \frac{\sum_p MD_{ip}\, E_{ip}}{G_i}. \tag{G.1}$$

This combines location-specific damages per ton, observed emissions intensity, and observed generation.

Region–fuel coefficients. I assign each plant to one of the thirteen model regions. For region j and fuel $k \in \{\text{coal}, \text{gas}\}$, the generation-weighted damage coefficient is

$$d_{jk} = \frac{\sum_{i \in (j,k)} G_i\, d_i}{\sum_{i \in (j,k)} G_i}. \tag{G.2}$$

The national generation-weighted rates are \$105/MWh for coal, \$11/MWh for existing gas, and \$4/MWh for recently built combined-cycle gas (AP4 2020 dollars). Table 14 reports the region-by-fuel coefficients.

Table 14: AP4 region-by-fuel local-pollution damage coefficients.

	Damage (\$/MWh, 2020\$)		Plants	
Region	Coal	Gas	Coal	Gas
CAL	—	5.8	0	60
CAR	74.3	6.2	14	26
CENT	96.2	3.6	21	69
FLA	84.9	10.4	7	51
MIDA	136.6	21.0	56	152
MIDW	123.5	12.8	98	227
NE	90.4	3.7	2	50
NW	32.4	3.7	30	67
NY	983.3†	10.1	3	66
SE	40.5	8.0	9	45
SW	41.4	6.2	8	34
TEN	109.1	2.8	6	4
TEX	126.4	9.9	19	122
National	105.3	11.0	273	973

Notes. Generation-weighted AP4 marginal damage per MWh by region and fuel, in 2020 dollars, before conversion to 2022 dollars. National rates are generation-weighted across all matched plants. †The New York coal coefficient is based on only three plants generating 0.9 TWh.

Application. A positive value is a damage avoided under the stable subsidy, since the unstable subsidy generates more fossil power. The coefficients are converted to 2022 dollars (Appendix G.4) and the annual stream is discounted at the model's 4% rate to 2010, giving $46 billion.

Table 15: Avoided local air-pollution mortality damages by source region.

Source region	Coal	Gas	Total
Midwest	23.7	1.3	24.9
Texas	11.8	0.4	12.2
Mid-Atlantic	0.0	2.5	2.5
Northwest	1.3	0.1	1.4
California	0.0	1.1	1.1
Central	0.8	0.1	0.9
Southwest	0.7	0.1	0.8
Carolinas	0.8	0.0	0.8
Florida	0.1	0.6	0.7
New York	0.0	0.4	0.4
New England	0.0	0.2	0.2
Southeast	0.1	0.1	0.1
Tennessee	0.0	0.0	0.0
National	**39.2**	**6.9**	**46.1**

Notes. Present value of avoided $PM_{2.5}$-related mortality damages (permanent minus budget-neutral uncertain), in billions of real 2022 dollars, discounted at 4% to 2010, by the source region of the emitting fossil plants. Damages are attributed to the region containing the emitting plants, not to the region whose residents are exposed; AP4 integrates exposure across the contiguous United States.

G.4 Welfare valuation, dollar years, and discounting

All monetary welfare values in the paper are expressed in real 2022 dollars, as a present value to 2010, over a horizon through 2070. The present-value base year of 2010 is the model's native reporting base; because the counterfactuals share the realized policy history through 2027, substantive differences between them begin only after the 2028 policy branch, up to numerical error in the stochastic baseline.

Consumption and resource cost. The model's real costs are calibrated in 2022 dollars. Consumption is discounted at the household rate implied by $\beta = 1/1.04$, so $PV_C = \sum_t \Delta C_t/(1.04)^{t-2010} = \$113.5$ billion.

Climate damages. I value avoided CO_2 with the EPA emissions-year-specific social cost of carbon under the 2% specification (U.S. Environmental Protection Agency, 2023). The EPA series is published in 2020 dollars; I convert the entire series once to 2022 dollars and do not inflate it forward year by year, because the EPA values are already emissions-year specific in constant dollars. With SCC_t the converted value and E_t emissions, $\text{ClimateBenefit}_t = (E_t^U - E_t^P)\, SCC_t$ and $PV_{climate} = \sum_t \text{ClimateBenefit}_t/(1.02)^{t-2010} = \$359.6$ billion.

Local air pollution. AP4 damages are in 2020 dollars, converted once to 2022 dollars. Unlike the social cost of carbon, AP4 reports contemporaneous monetized mortality damages from a given year's emissions and does not embed a future stream of damages. I discount the annual local-pollution stream at the model's private 4% rate as a modeling convention for contemporaneous monetized damages, giving $PV_{local} = \$46.1$ billion.

Inflation. External monetary estimates are converted to 2022 dollars with the BEA GDP implicit price deflator (U.S. Bureau of Economic Analysis, 2024): the annual index rises from 105.4 in 2020 to 118.0 in 2022, a factor of 1.120. Changing the dollar year is not an economic change; it places consumption, climate damages, and local-pollution damages in common units.

Discount rates. The components use different rates by design: consumption at the private rate implied by $\beta = 1/1.04$; climate damages are valued using the EPA's year-specific social cost of carbon under its 2% discount-rate scenario; and local mortality damages at the private rate. I do not impose a single social discount rate.

Table 16: Welfare valuation conventions.

Component	Dollar year	PV year	Discount rate
Consumption / resource cost	2022$	2010	4%
CO_2 damages	2022$	2010	2% (EPA)
Local mortality damages	2022$	2010	4%

Table 17: Value of continuous support across forward regimes (present value, SCC = \$213/tCO$_2$ in 2022 dollars).

Permanent subsidy vs.	Climate benefit (\$B)	Consumption cost (\$B, PV)	Local air pollution (\$B)	Value (compreh.) (\$B)	Excl. local air poll. (\$B)	Breakeven SCC (\$/tCO$_2$)
Budget-neutral uncertain subsidy	360	113	46	+292	+246	67
No subsidy	563	247	70	+385	+315	94
Cross-party	53	5	8	+56	+48	21

Notes. As Table 5, for the forward regimes. The comprehensive value adds avoided local air-pollution ($PM_{2.5}$) mortality damages (AP4, discounted at 4%) to the climate-only net value; the *Excl. local air poll.* column reports the climate-only net value. The breakeven social cost of carbon in the last column is computed on the climate channel. *No subsidy* is the permanent repeal. The value is positive and the breakeven lies far below the EPA's \$213 in every regime.

H Robustness

The value of a stable subsidy reported in this appendix is comprehensive, including both avoided climate damages and avoided local air-pollution mortality damages; the tables also report the climate-only value, excluding local air pollution.

H.1 Policy persistence

The expected duration of a lapse is the main uncertainty in the policy calibration. I compare the baseline with a faster renewal regime, in which the subsidy is active 85% of the time in the long run, and with permanent repeal. Table 17 reports the results.

H.2 Climate discount rate

In the baseline, private consumption is discounted at the model's market rate while climate damages are discounted at the social rate used in the EPA SCC calculation. Table 18 varies the rate applied to climate damages. Even when climate damages are discounted at the market rate, the value of a stable subsidy remains positive.

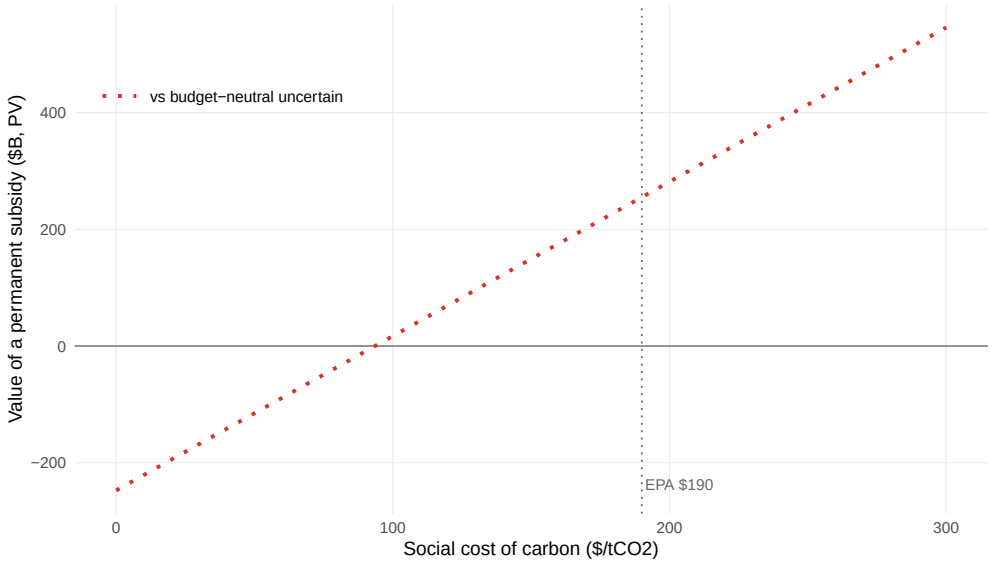


Figure 16: Value of a stable subsidy and the social cost of carbon.
Notes. Welfare difference between the permanent and budget-neutral uncertain subsidies as the social cost of carbon varies.

Table 18: Welfare results across the climate discount rate.

	Value of continuous support (permanent − uncertain)			Value of the subsidy (uncertain − no subsidy)		
Climate rate	Climate ($B)	Value ($B)	Breakeven SCC	Climate ($B)	Value ($B)	Breakeven SCC
1.5%	433	+320	56	247	+113	115
2.0% (baseline)	360	+246	67	203	+69	140
3.0%	251	+137	96	139	+5	206
4.5% (market)	150	+37	161	81	−53	354

Notes. The value of continuous support over the budget-neutral uncertain subsidy (columns 1–3) and of the uncertain subsidy over no subsidy (columns 4–6), as the rate used to discount the climate benefit varies. Only the climate benefit is re-discounted; the consumption cost is held at its present value at the model rate ($113 billion for continuous support, $134 billion for the subsidy), since the discount factor β is a structural parameter. The baseline is the EPA's 2% social rate; the 4.5% row discounts the climate benefit at the model market rate, so the climate externality is treated like private consumption. Value = climate − consumption cost; breakeven SCC is the social cost of carbon at which the value is zero (the reference path is scaled from the EPA central value, $\$213/tCO_2$ in 2022 dollars). The value of continuous support stays positive at every rate; the value of the subsidy turns negative once the climate benefit is discounted at the market rate.

H.3 Reporting horizon and single-rate discounting

The headline truncates avoided emissions at 2070 while discounting the consumption cost over the model's full horizon, so it understates the infinite-horizon value. A related question is whether climate damages and consumption should share a single discount rate, as is standard in macroeconomics, rather than the baseline's dual rates (the social rate for climate, the market rate for consumption). Table 19 reports the climate-only value of a stable subsidy across both dimensions; it is positive in every cell.

Table 19: Value of a stable subsidy: climate discount rate and reporting horizon.

Climate discount rate	Emissions to 2070 ($B)	Infinite horizon ($B)
2% (EPA social rate, baseline)	246	590
4% (model market rate)	65	119

Notes. Climate-only value of a stable subsidy (versus the budget-neutral unstable subsidy), present value in billions of 2022 dollars. The consumption cost ($113 billion) is measured over the model's full horizon in every cell; only the rate applied to avoided CO_2 and the emissions-reporting horizon vary. The infinite-horizon column extends the terminal steady-state abated-emissions flow forward, discounted at the stated rate, matching the treatment of the consumption tail. Extending the explicit transition to 2130 leaves the capacity and emissions gaps open rather than closing them, so the horizon is a reporting choice, not a convergence artifact.

Under the fully consistent macroeconomic convention, discounting climate damages, local air pollution, and consumption all at the model's market rate over an infinite horizon, the stable subsidy remains valuable: the comprehensive breakeven social cost of carbon is $\$88/tCO_2$, below the EPA central estimate of $190, and at the social rate the avoided local air-pollution mortality alone exceeds the consumption cost. The result is thus robust to a single market discount rate; it would be overturned only by combining the market rate with the pre-2023 low social-cost-of-carbon values that the EPA's 2023 update superseded.

H.4 Fuel prices

The counterfactual holds the natural-gas price path at its EIA reference level. Cheaper gas makes renewables less competitive without the subsidy, while dearer gas makes them more competitive, so the value of a stable subsidy could move in either direction. I re-solve the model with the gas price scaled by 0.65 (low, roughly the EIA high-supply case) and by 1.40 (high, low-supply), rematching the budget in each case. The comprehensive value of a stable subsidy is larger in both: $534 billion under low gas prices and $382 billion under

high gas prices, against $292 billion in the baseline. Under low gas prices the uncertain economy leans further on gas, widening the emissions gap; under high gas prices the fossil-reliant uncertain path pays higher fuel bills, so the stable subsidy costs almost nothing in consumption. The reference gas path is therefore a conservative middle.

H.5 Renewable technology costs

The baseline wind and solar capital-cost paths decline more slowly than any NREL ATB 2024 scenario. As a robustness I adopt the ATB *conservative* trajectory, the scenario assuming the smallest cost reductions, which nonetheless declines faster than the baseline (a 21% wind and 43% solar reduction from 2024 to 2050, against the baseline's 17% and 22%); I rematch the budget. Cheaper renewables might be expected to make the subsidy less pivotal, but instead they widen the gap between the stable-subsidy and unstable-subsidy economies: under the stable subsidy the economy exploits the lower costs and builds more wind, while the uncertain economy still delays investment during lapses. The comprehensive value of a stable subsidy rises to $326 billion ($274 billion excluding local air pollution), against $292 billion in the baseline. The result is therefore robust to, and if anything strengthened by, lower renewable capital costs.

H.6 Perfect-foresight investment by technology

Figure 17 separates the renewable-investment response in Section 7.3 into wind and solar. The sharp anticipatory response is concentrated in wind, for which eligibility determines the stream of production-tax-credit payments associated with newly qualifying capacity.

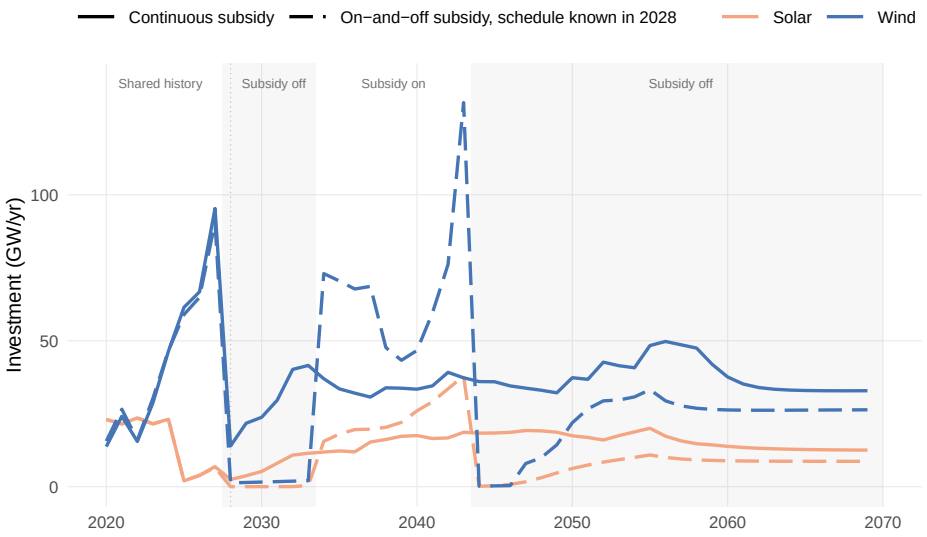


Figure 17: Wind and solar investment under perfect foresight.

Notes. Annual wind and solar investment under the stable subsidy and under the switching schedule that firms learn in 2028 (the short-lapse schedule of Figure 10). Shaded areas mark the years in which the subsidy is off under that schedule.

I Additional counterfactual results

I.1 Regional results

Figure 18 reports the stable-minus-unstable capacity gaps and unstable-minus-stable emissions gap for all 13 regions. The largest wind-capacity differences occur in the interior regions where wind resources are strong and the federal subsidy affects the marginal investment decision. The gas-capacity and emissions responses need not occur in the same regions as the renewable-investment response because electricity is traded within interconnections.

Table 20 reports the regional decomposition of the value of the unstable subsidy relative to no subsidy.

Table 20: The value of the uncertain subsidy by region, decomposed (uncertain versus no subsidy).

	Climate gains				Local poll.†	Consumption					
Region	Coal	Ex. gas	New gas	Total	Avoided	Invest.	Income	Wages	Profit	Total	Value
Midwest	56	18	7	81	15	−45	29	0	0	−15	66
California	0	22	2	24	0	−16	6	2	−3	−11	13
Mid-Atlantic	0	15	3	18	2	−10	5	0	0	−6	12
Florida	2	6	0	7	0	−5	2	0	0	−3	4
New England	0	9	1	10	0	−9	4	1	−2	−6	4
Carolinas	5	0	0	5	1	−2	0	0	0	−2	3
New York	0	7	0	7	0	−6	3	1	−1	−4	3
Southeast	1	1	0	2	0	−1	0	0	0	−1	1
Texas	15	4	0	19	4	−49	34	−4	1	−18	1
Southwest	3	2	0	6	0	−6	−2	1	0	−7	−2
Tennessee	0	3	0	3	0	−11	1	0	0	−10	−7
Northwest	14	−2	1	14	1	−20	−8	7	−3	−24	−10
Central	0	2	5	6	0	−23	−7	3	0	−27	−21
National	**95**	**63**	**45**	**203**	**24**	**−203**	**66**	**12**	**−9**	**−134**	**69**

Notes. Present value in billions of dollars, by region, sorted by value. Climate gains value the region's own avoided CO_2 at the EPA year-specific social cost of carbon discounted at 2%, split by source (existing coal, existing gas, new gas). Consumption is the change in the present value of the region's consumption; income, wages, and profit raise it, generation investment enters negatively. Value = climate gains + consumption. Rows sum to the National row. †The final column reports avoided local air-pollution ($PM_{2.5}$) mortality damages attributed to emissions from fossil plants *located in* the region (AP4, discounted at 4%); it is shown for reference, is *not* added to Value, and is not a measure of health benefits received by residents of the region.

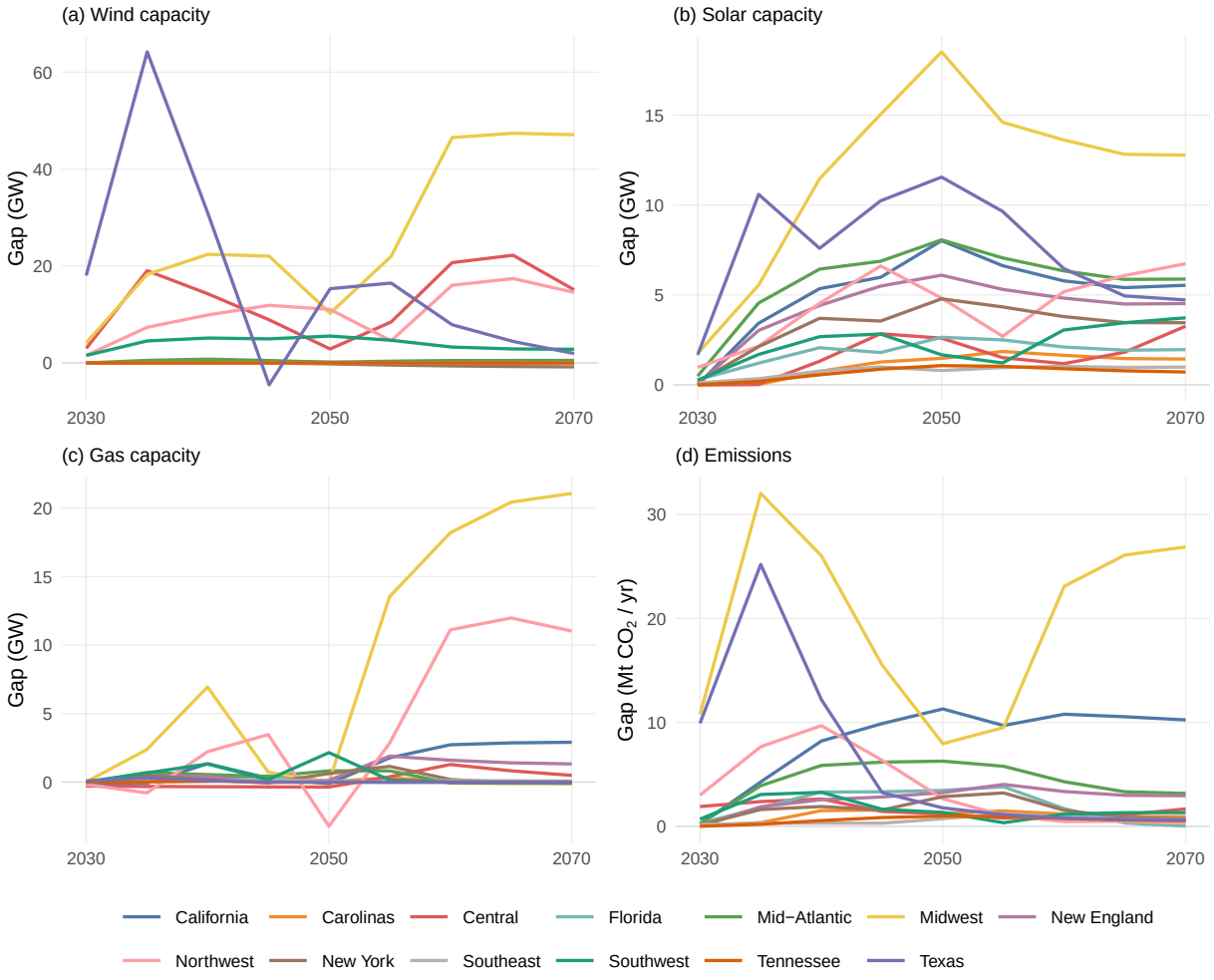


Figure 18: Regional gaps over time, all thirteen regions.

Notes. Gaps between the stable subsidy and the budget-neutral unstable subsidy at five-year intervals. Panels (a)–(b) report wind and solar capacity as stable minus unstable. Panels (c)–(d) report gas capacity and annual CO_2 emissions as unstable minus stable.